

COMPARATIVE PHYLOGEOGRAPHY AND MOLECULAR EVOLUTION OF AUSTRALIAN EUGONGYLINAE SKINKS



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i3L – Indonesia International Institute for Life Sciences
School of Life Sciences

In partial fulfillment of the requirements for the Bachelor of Science in
Bio Informatics

Jakarta, Indonesia
2020

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"Comparative Phylogeography and Molecular Evolution of Australian Eugongylinae Skinks"

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Abstract

Temperature variation over time and space has shaped the population structure and diversity, as well as molecular evolution of species. In order to assess this idea, we utilized a large exon-capture sequence data of Australian Eugongylinae skinks as examples of temperature-sensitive ectotherms. The effect of temperature in the late Quaternary is assessed based on nDNA data of southern *Morethia* skinks (a member of Eugongylinae subfamily) using custom bioinformatics pipelines (sequence assembly), IQTree (phylogenetic tree), PopGenome (population diversity), *Structure*, *conStruct* (population structure), and BEAST2 (population size). On the other hand, the association between temperature (and other factors) on long-term substitution rates was assessed based on a large exonic sequence of 44 species pairs from Eugongylinae subfamily using Poisson regression analysis. The results show that species with narrower realized climate niche (*M. obscura*, *M. adalaidensis*) have very weak phylogeographic structure with evidence of rapid population growth post-LGM, while other species persisted in multiple refugia during LGM. The aridification histories of northern and southern Australia also affected the timing of population expansion, with southern *M. obscura* experienced more recent radiation (~0.3Mya) than northern monsoonal *M. ruficauda* (~2Mya). At macroevolutionary scale, temperature is also the strongest, positively-correlated predictor of variation in synonymous and non-synonymous substitution rates across the Australian Eugongylinae. It explains 35% and 40% of variation in dN and dS, respectively, and about 20% of variation in both after controlling for other factors. Overall, this studies points to temperature as a strong driver of both micro- and macro-evolutionary processes in these lizards.

Acknowledgements

In the completion of this project, I first want to thank God for the opportunity that He gave me to do my Thesis's project at Australian National University. Secondly, I want to thank my parents who supported me throughout this period, both materially and emotionally.

I also want to express my gratitude for Prof. Craig Moritz who welcomed me in his group, as well as Dr. Sally Potter and Dr. Xia Hua who patiently taught and guided me throughout the projects. Special thanks for Mr. Rustamzhon Turakulov from ANU Research School of Biology, Mr. David Duchene from ANU Research School of Biology, Prof. Simon Ho from University of Sidney, Mr. Mozes Blom from Swedish Museum of Natural History, Ms. Ana Catarina Silva, and everyone else who helped me along the way.

Lastly, I want to thank Mr. Rizky Nurdiansyah who helped and gave me ideas since the preparation of this project until the drafting of Thesis report. May the endless grace of God bless all of you and everything that you do.

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List of Abbreviations

AHE	anchored hybrid enrichment
ALA	Atlas of Living Australia
AMT	annual mean temperature
AMT	Australian Monsoonal Tropics
AWT	Australian Wet Tropics
AZ	arid zone
bp	base pairs
BWA	Burrows-Wheeler Aligner
dN	non-synonymous mutation rate
DNA	deoxyribonucleic acid
dS	synonymous mutation rate
EAPHy	Exon Alignment for Phylogenetics
EBSP	extended bayesian skyline plot
ESH	evolutionary speed hypothesis
GATK	Genome Analysis Toolkit
GTR	Generalised Time Reversible
HKY	Hasegawa, Kishino, and Yano (model)
IBD	isolation by distance
IUPAC	International Union of Pure and Applied Chemistry
kya	thousand years ago
LGM	last glacial maximum
MCMC	Markov Chain Monte Carlo
MG	Muse and Gaut (model)
ML	maximum-likelihood
MRCA	most recent common ancestor
mtDNA	mitochondrial DNA
Mya	million years ago
ND4	NADH-Ubiquinone Oxidoreductase Core Subunit 4

nDNA	nuclear DNA
O(Het)	observed heterozygosity
OMG	Oz Mammals Genomics
ORF	open reading frame
PCoA	principal coordinate analysis
PCR	polymerase chain reaction
RNA	ribonucleic acid
RNASeq	RNA sequencing
RTG	Real Time Genomics
SNP	single nucleotide polymorphism
SW	South-West
TCCP	Target Cloud Capture Platform

Chapter I. Introduction

Temperature undeniably plays a vital role in the life of organisms. For instance, organismal growth, metabolism, and even reproduction are affected by changes in temperature (e.g. Wudarski et al., 2019). Thus, it is important to understand the interaction between temperature and living things. In this study, our main aim is to assess how does climatic variation over time and space influence the evolution and diversity of organisms. In order to answer this question, we conducted two projects that investigate evolutionary responses at two different scales: (i) effects of late Quaternary climatic variation on population and genetic diversity within species, as well as (ii) effects of temperature (and other factors) on long-term rates of molecular evolution.

For both projects, we focus on the radiation of *Eugongylinae* skinks across Australia, analyzing large scale (>1000 loci) of exon-capture sequence data. Skinks, as temperature-sensitive ectotherms, are expected to be sensitive to temperature variation (Huey, 1982). Exonic sequences of the respective group allow for micro- to macro-evolutionary analyses, as well as comparison of mutations with little or no expected effect (synonymous) versus those that could affect protein structure and function (non-synonymous).

1.1. Population structure and phylogenetics of southern *Morethia* skinks

Australia is the driest continent in the world after Antarctica: seventy percent of the country receives less than 500 mm of rain per year, while half of that number is considered as desert (Geoscience Australia, no date). Even though it constitutes a large portion of the world's arid area, Australian desert is considered young. According to Byrne et al. (2008), the evidence of aridity in Australia started during the mid-Miocene, denoted by cessation of major depositional systems. While the early Pliocene introduced a warmer and wetter climate, the following glacial - interglacial cycles during the Pleistocene period resulted in the expansion of arid area around 2.5 million years ago. As a climax, the Last Glacial Maximum (LGM) introduced widespread hyper-aridity, causing

most of the area to become inhospitable for many organisms. As a result, such organisms were forced to contract to mesic refugia or adapt to the new environments.

Organisms that share common environmental niches will interact with each other, forming biological communities within biomes, as defined by climatic and broad vegetation types (Purves, Sadava, Orians & Heller, 2003). As each climatically-defined biome should undergo different environmental histories, related taxa that live in different biomes, or parts thereof, might have responded differently to the climatic changes during the last Quaternary Period, especially through the LGM and following Holocene (Byrne et al., 2008). On average, species with more restricted geographic and climatic ranges should be more specialised physiologically and thus more sensitive to abrupt of large-scale shifts in climate. If they had not already lived in the refugium, the species would suffer the most during the harsh climatic periods as it becomes harder for them to find a suitable habitat. This is likely to result in strong contraction during the cold-dry LGM, followed by major expansion once the climate becomes more friendly and the habitats turn more hospitable. In contrary, species with broader climate tolerances could show greater persistence across their ranges, thus maintaining phylogeographic structure with only localized expansions (Hewitt, 1996). Organismal pattern of contraction and the following expansion can be determined by assessing the phylogenetics and population genetics of the organisms. If the species' ranges contracted into one refugium, the population should have a strong signature of expansion with little phylogeographic structure. In contrast, persistence across the range (or in multiple dispersed refugia), would result in either weak or strong phylogeographic structure and less evidence of species-wide population expansion (Hahn, 2018).

In this project, we utilized the exonic sequences from southern temperate or arid-zone *Morethia* skinks, aiming to analyze the phylogeographic structure and demographic history of each species and compare these with analogous results for their northern (arid or monsoonal savanna) relatives reported by Potter et al. (2019).

1.2. Molecular evolution of Australian Eugongylinae skinks

Australia has multiple biomes, each with its own climatic pattern and history (Byrne, 2008; Fitzsimmons et al., 2013). This results in a gradient of temperature means from the hotter (or more variable) northern Australia to the more temperate southern region (Atlas of Living Australia, no date), which in turn is predicted to affect species molecular evolution rate. According to the evolutionary speed hypothesis (ESH), higher temperatures, such as in the tropics, lead to faster evolutionary rates due to increased mutation rate (Rohde, 1992; Oppold et al., 2016; Orton, May, Ly, Lee & Adamowicz, 2018). This theory also highlights the role of life-history traits, including generation time, in relation with the rates of synonymous (dS) and non-synonymous mutation (dN). However, it still requires empirical testing, particularly for Squamates (Brown, 2013).

In order to answer this problem, most studies have used as their primary data mitochondrial DNA (mtDNA), which has very high mutation rate compared to nuclear DNA (nDNA) (Allio, Donega, Galtier & Nabholz, 2017) and represents just one linked gene. For a broader perspective, we used a large set of sequences from nuclear exons (~1000 exons) from the entire Eugongylinae subfamily (of which *Morethia* is a part); this is similar with approaches from Garcia-Porta et al. (2019) that utilized nDNA data from lacertid lizards. We included twenty-one genera from the Eugongylus group, aiming to assess the correlation between mutation rate and mean temperature and/or life-history traits of the species.

Chapter II. Literature Review

2.1. Data set: Australian Eugongylinae skinks

Eugongylinae is a subfamily of skinks that are found throughout Australia from the hot tropics to the coldest, southern-most species in south of Tasmania and having diverse ecological and physiological traits (Brandley et al., 2015). The members of this group range from large radiations of mostly tropical genera (e.g. *Carlia*; Bragg et al., 2018) or continental-scale radiations (*Cryptoblepharus*; Blom, Horner & Moritz, 2016) to temperate radiation (e.g. *Niveoscincus*; Hutchinson & Schwaner, 1991) and relictual monotypic genera, mostly in mesic east coast forests (e.g. *Pseudemoia*; Hutchinson & Donnellan, 1992). This includes *Morethia*, an endemic Australian genus of skink that inhabits a wide range of biomes, ranging from the monsoonal tropics, eastern woodlands, northern, and southern deserts (Greer, 1980; Cogger, 2014; Wilson & Swan, 2017). They are small, terrestrial, diurnal skinks that can be differentiated based on the pattern and nature of suture-line between their supra-ciliary and supra-ocular scales. Up until now, eight *Morethia* species have been identified: *M. adalaidensis*, *M. boulengeri*, *M. butleri*, *M. lineocellata*, *M. obscura*, *M. ruficauda*, *M. storri*, and *M. taeniopleura*. While most of the species live in woodland and semi-arid to arid shrubland (with the exception of *M. adalaidensis*, which is almost restricted to the chenopod shrubland, a largely treeless vegetation type in arid southern Australia, including the margins in salt lakes), some species are also found in dry forest (*M. boulengeri*), heathland (*M. lineocellata*), stony hills and ranges (*M. ruficauda*, *M. taeniopleura*), and grassland (*M. storri*) (Greer, 1980; Cogger, 2014).

2.2. Effects of late Quaternary climatic changes on population structure and diversity in Australia

In general, Australia comprised of multiple biomes: Australian Wet Tropics (AWT), temperate (mesic) regions in the east and south, arid zone (AZ), and Australian Monsoonal Tropics (AMT) (Figure 1). Each of them has different climatic features, significantly affected by two major

weather systems: monsoons and westerlies. The northern AMT and AZ are influenced by the Asian monsoon, and the AWT by South-East Trades, resulting in summer rainfall and a dry winter. In contrary, southern Australia, that consists of the central AZ and mesic, temperate regions along the east coast and in the south-west, are controlled by the southern westerly systems, causing dry, hot summer and rainfall winter (Byrne, 2008; Fitzsimmons et al., 2013).

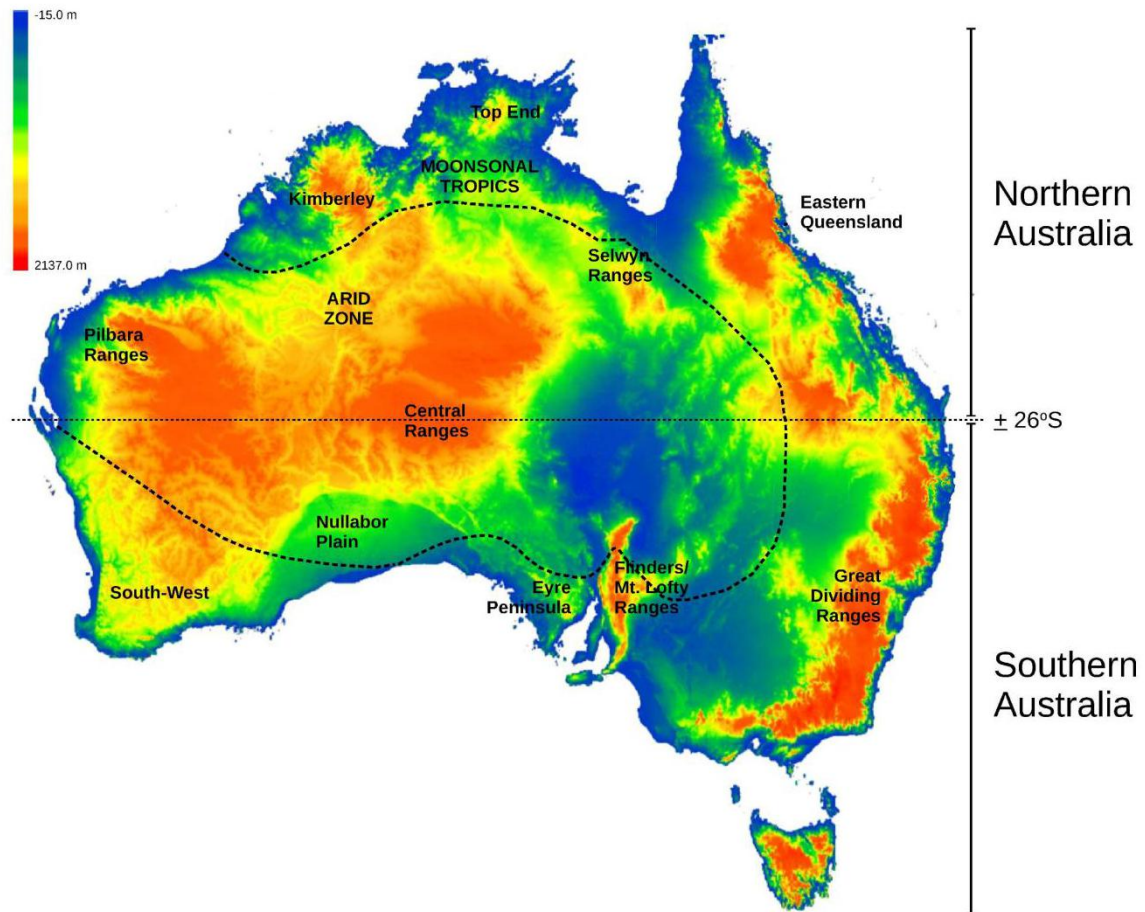


Figure 1. Elevation map of Australia showing major biomes and refugia mentioned in the text. The dashed line shows the arid zone boundary. Elevation data is derived from Atlas of Living Australia (ALA; Atlas of Living Australia, no date).

While each biome has different history (Schneider, Cunningham & Moritz, 1998; Byrne et al., 2008; Bowman et al., 2010; Byrne et al., 2011; Potter et al., 2019), the northern and southern arid regions also underwent different environmental changes in the last 40 kya (Fitzsimmons et al., 2013). After the humid conditions between ~40-30 kya, increase of dune activity and transport reflected the increased aridity across the continent, peaking at the LGM around 23-18 kya. After the LGM

ended, the story differed: the westerly-influenced southern AZ experienced immediate precipitation increase in ~18-14 kya, followed by an increasing aridity until ~10 kya and, after the mesic early Holocene, a return to arid conditions in the late Holocene. On the other hand, monsoon-controlled northern AZ remained arid until ~14 kya when the monsoon strengthened and moved southwards, and experienced less severe aridity from the mid Holocene to the present. In addition to the contrast between northern and southern AZ, there is also evidence of strong flows in the Murray Basin rivers through the eastern AZ from before the LGM to the Holocene, likely due to runoff from the eastern highlands (Hesse, Magee & van der Kaars, 2004; Fitzsimmons et al., 2013). This could have retained riparian vegetation in an otherwise sparsely vegetated region. The contrasting climate histories between northern and southern Australian AZ, eastern and western AZ, as well as AMT and arid zone, is expected to result in differing scales of change in population size and geographic range for otherwise similar species.

In addition to multiple biomes, there are several known geographical barriers and refugia in Australia (Figure 1). Mountain ranges in the AZ, such as Pilbara (Hamersley), Central, and Selwyn Ranges, have been identified as important refugia for organisms (e.g. Pepper, Doughty & Keogh, 2013; Noble, Laver, Rosauer, Ferrier & Moritz, 2017; Byrne et al., 2008), accompanied by more mesic regions around the Kimberley and Top End in northern Australia (Potter, Eldridge, Taggart & Cooper, 2012; Potter et al., 2019) as well as South-West and Flinders Ranges in the south (Keast, 1961; Ford, 1987). These areas should serve as reservoirs for species during the harsh condition. The Nullarbor Plain has also specifically been identified as geographical barrier between the temperate mesic zones in south-west and south-east regions (Byrne et al., 2008; Ansari et al., 2019).

Distribution and climatic range of Morethia skinks

A previous study (Potter et al., 2019) showed that northern *Morethia* species had varied responses to past climate change, with strongly-structured lineages and local persistence in the monsoonal tropics, especially for *M. storri*, and evidence of range expansion and lower diversity

across the northern desert range of *M. ruficauda*. MaxEnt (Maximum Entropy) species distribution models (Elith et al., 2010) of species of *Morethia*, an endemic genus of skinks in Australia, based on the distribution data from the ALA (Atlas of Living Australia, no date) with default settings is shown in Figure 2. Four BioClim variables are considered: annual precipitation (Bio12), annual precipitation seasonality (Bio15), annual temperature mean (Bio01), and annual temperature seasonality (Bio04), accounting for water availability and temperature of the region. The models generated south-west area as the most suitable niche for Australian *M. obscura* (Figure 2E), south-east area for *M. boulengeri* (Figure 2B), multiple highly suitable areas for both *M. adalaidensis* and *M. butleri* (Figure 2A, C), as well as coastal area for *M. lineoocellata* (Figure 2D) and *M. taeniopleura* (Figure 2H). However, the results are mixed. In *M. obscura* and *M. butleri*, annual mean temperature accounts for more than 50% to the models, inferring a strong correlation between species distribution with temperature (compared to water availability). This is different to other *Morethia* species (i.e. *M. adalaidensis*, *M. boulengeri*, *M. lineoocellata*, *M. taeniopleura*), where annual precipitation and seasonality contribute the most. Between the northern species, *M. ruficauda* has multiple areas of high suitability which are mostly affected by precipitation seasonality (Figure 2F; 50.6%); this reflected the wide habitat distribution of *M. ruficauda* which comprised arid zone and mesic monsoonal tropics area. On the other hand, *M. storri* had limited niche around AMT with annual mean temperature as the most significant factor (Figure 2G; 72.5%). Figure 2 further shows that several species have wider ranges of habitat (e.g. *M. boulengeri*) compared to others (*M. lineoocellata* and *M. obscura*).

The ecological and climatic limitation of the species should affect the contraction and expansion patterns of the species in response to climate change. Thus, species with more restricted climatic tolerances or habitat might show great signature of expansion following the harsh periods, denoted by significant increase in the value of effective population size across time. We hypothesized that there would be: (i) greater local persistence in eastern than western AZ taxa due to the presence of lakes and rivers through the hyper-arid LGM in the eastern AZ providing more

persistent suitable habitat for the species; (ii) different extent/timing of population expansions between southern and northern AZ due to difference in timing of historical aridification events; and (iii) stronger fluctuations in taxa with narrower realized climate niches (i.e. climate tolerance and/or habitat), including *M. adalaidensis* and *M. obscura* (see Results).

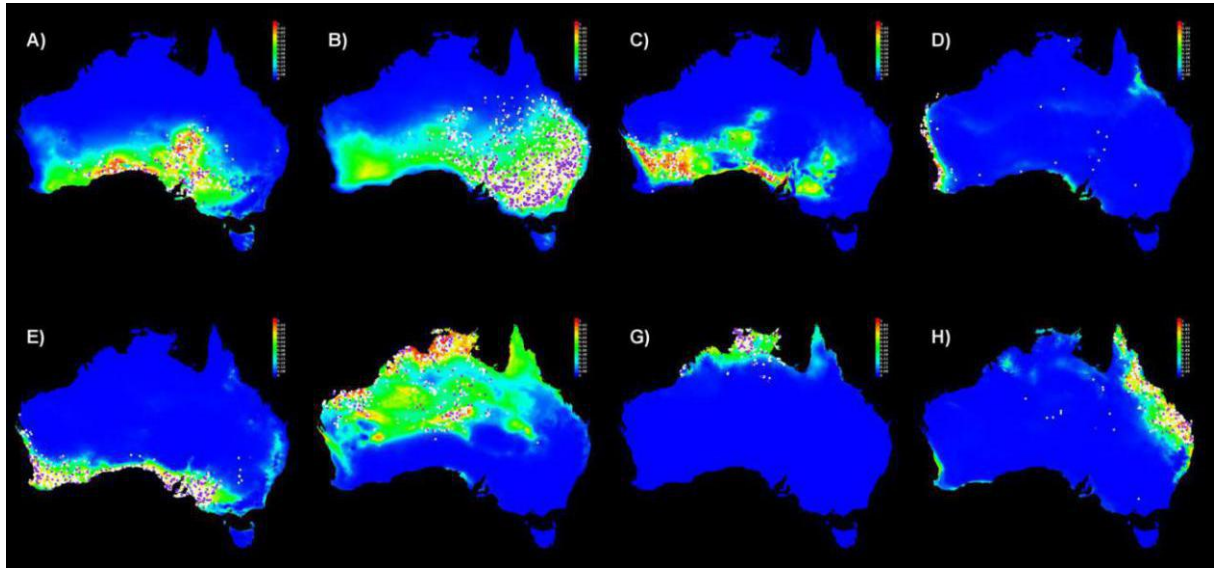


Figure 2. Maxent model for Australian *Morethia* species. A: *M. adalaidensis*, B: *M. boulengeri*, C: *M. butleri*, D: *M. lineocellata*, E: *M. obscura*, F: *M. ruficauda*, G: *M. storri*, and H: *M. taeniopleura*. The model derived from Atlas of Living Australia Spatial Portal (Atlas of Living Australia, no date).

2.3. Effects of temperature variation on molecular evolution rates

The fitness of ectotherms, including skinks, is sensitive to environmental temperature (Deutsch et al., 2008; Huey et al., 2012; Sinervo et al., 2010). This becomes particularly apparent in species that live in the tropics. Thus, further increase in environmental temperatures due to global warming might exceed the organism's thermal (physiological) optimum, resulting in reduced fitness and ultimately extinction. Hua & Bromham (2017) further argued that mutation rate is very responsive to environmental pressures. The association itself might be direct (i.e. increase in temperature induces faster rates of biochemical reaction, resulting in more free radicals that can cause DNA damage and thus mutation), indirect (i.e. higher temperature results in faster life history, increasing the risk of copy errors), or both. Based on this idea, species that live in warmer environments should have higher mutation rate (and, under neutral theory, substitution rate) than

the ones in more temperate habitats, denoted by longer relative phylogenetic branches. However, the empirical support is still decidedly mixed. Previous studies showed that the predictions are true for a large number of ectothermic organisms, including plants (Gillman, Keeling, Gardner & Wright, 2010), fishes (Wright, Ross, Jeanette Keeling, McBride & Gillman, 2010), and amphibians (Wright, Gillman, Ross & Keeling, 2010). While the theory seems to be universally applied for cold-blooded organisms, another study, focused mostly on mtDNA, found that it was not the case for Squamata (Rolland, Loiseau, Romiguier & Salamin, 2016). They found no significant correlation between temperature and branch length, while the diversity of squamates itself peaked in mid-latitude regions (i.e. 20 degree of latitude) instead of the Equator. In contrary, Garcia-Porta et al. (2019) found statistically-significant positive association between temperature and substitution rate among lacertid lizards based on a large set of nuclear loci derived from a combination of RNASeq and anchored hybrid enrichment sequencing (AHE) data.

According to the evolutionary theory of senescence, species with low mortality pressure (e.g. predation, temperature, competition) will have longer lifespan due to the selection pressure against deleterious mutations (Stark et al., 2018). This means that life-history traits, such as generation time and body size, might also be associated with mutation rate (Bromham, 2011; Hua & Bromham, 2017). For example, species that live in tropics will have shorter generation time than the ones in temperate habitats due to higher mortality pressure and metabolic rates. This results in higher mutation rate due to more frequent DNA replication that is prone to error. As the traits strongly correlate with each other (e.g. species with longer generation time usually have bigger body size), it is important to consider their covariations in the analysis.

In this study, we utilized exon-capture data of the entire *Eugongylus* group, which spans the full range of temperature conditions across the continent. Thus, it allows us to better evaluate the correlation between temperature and molecular evolution rate. In addition, we will incorporate life-history traits of each species to assess their roles in the correlation. We hypothesized that temperature is the main driver of molecular evolution, where the increase in temperature will

increase the synonymous and non-synonymous mutation rates of the species due to aforementioned explanation.

Chapter III. Materials and Methods

3.1. Population structure and phylogenetics of southern *Morethia* skinks

This part utilizes similar approaches to Potter et al. (2019) so as to allow for direct comparison of results. The exon-capture sequence data from southern *Morethia* species were assembled using Oz Mammals Genomics (OMG) pipeline. Then, the resulting Variant Call Format (VCF) and haplotypes files were used to infer the observed heterozygosity and depth of each sample, as well as population structure and diversity, respectively. The complete flowchart for processing the genomic data is shown in Figure 3, while the sampling maps of each southern *Morethia* species in Figure 4 [colorization of samples reflects major lineages from the mtDNA tree (see Figure 7; left)].

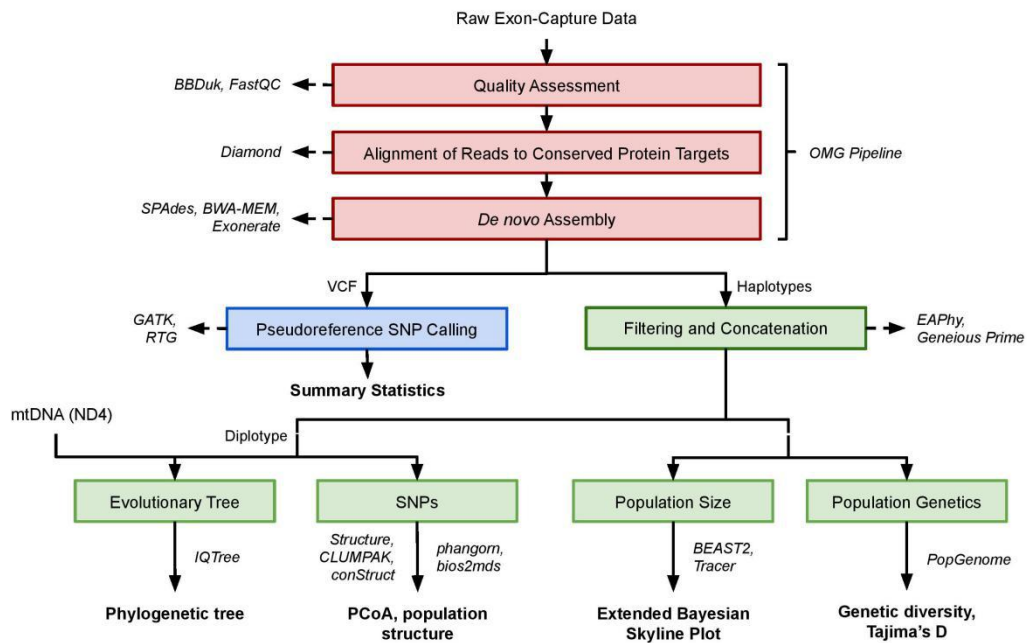


Figure 3. Pipeline of the phylogeographic structure analysis. Red: OMG/TCCP pipeline; blue: summary statistics analysis; green: phylogenetics and population genetics analysis. The complete lists of software that were utilized can be found in the paragraph below.

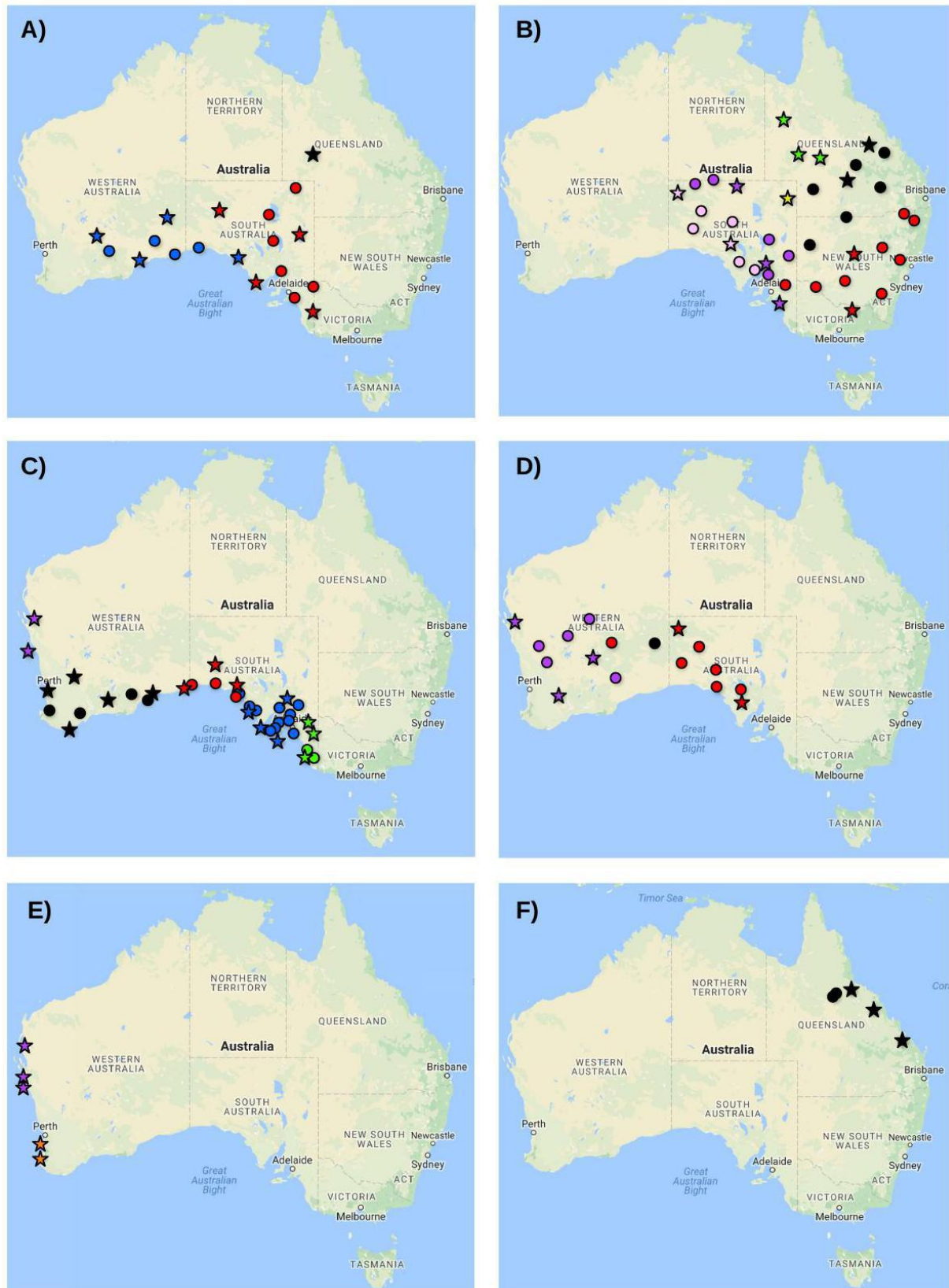


Figure 4. Sampling map of southern *Morethia* species: *M. adelaidensis* (A), *M. boulengeri* (B), *M. obscura* (C), *M. butleri* (D), *M. lineocellata* (E), *M. taeniopleura* (F). Circles: mtDNA-only samples; stars: mtDNA samples that further sequenced for exons. Samples were colored based on major mtDNA lineages (see Figure 7; left).

3.1.1. Genomic data set

The data set comprised of mtDNA and nDNA sequences. The mtDNA-ND4 data comes from 120 southern *Morethia* samples (Figure 4), which were selected to span the geographic ranges of each species where samples were available: *M. lineocellata* (5 samples), *M. obscura* (38 samples), *M. adalaidensis* (19 samples), *M. taeniopleura* (5 samples), *M. boulengeri* (38 samples), *M. butleri* (15 samples). One sample from *Proablepharus kinghorni* (currently known as *Australoblepharus kinghorni*) was used as an outgroup. Then, sub-sampling was done for exon-capture sequencing by considering the span of geographic and mtDNA diversity. As we focus on AZ taxa, there is limited sampling of coastal species (*M. lineocellata* and *M. taeniopleura*).

The exon-capture data was generated prior to my study from 53 southern *Morethia* samples to complement data from northern species and following the same laboratory methods (Potter et al, 2019): *M. lineocellata* (5 samples), *M. obscura* (18 samples), *M. adalaidensis* (9 samples), *M. taeniopleura* (3 samples), *M. boulengeri* (13 samples), *M. butleri* (5 samples) (Figure 4). Exome-capture sequencing is a cost-effective tool to identify genetic variation within coding regions (Puritz & Lotterhos, 2018). It implements a target-enrichment strategy to “capture” sequences from exonic regions of the genome via hybridization with synthetic oligonucleotide probes with complimentary sequences. In this study, the probes were designed based on orthologous loci from *Eugongylus* group transcriptome data (Bragg et al. 2016) , allowing us to utilize the probes for other distantly-related organisms where data is unavailable. The exon-enriched DNA library for each sample is later sequenced via high-throughput sequencing platforms. Exon-capture has become popular due to its sensitivity in identifying variants that cause phenotype alterations with much lower cost than whole-genome sequencing (Puritz & Lotterhos, 2018; Radio et al., 2019). Moreover, it allows the separation of synonymous (silent) and non-synonymous changes, with the former being subjected to selection. A further advantage of exon-capture is that it yields highly complete data for the same (orthologous) loci across taxa. This information enables robust estimation of phylogeographic structure and population diversity as well as tests for population expansion (e.g. Potter et al., 2018; 2019).

3.1.2. *De novo* assembly

Exon capture data were assembled based on the OMG pipeline (Bragg et al, 2016; Oz Mammals Genomics, 2019). First, the raw sequences were trimmed, filtered, and masked by using BBDuk from BBtools v37 (Bushnell, Rood & Singer, 2017) based on the following parameters: minimum read-length (*minlen*) of 35, kmer-length (*k*) of 27, minimum average quality (*trimq*) of 20, minimum kmer-length (*mink*) of 11, and maximum Hamming distance (*hdist*) of one. The sequencing qualities of the filtered sequences were assessed by using FastQC v0.11.5 (Babraham Bioinformatics, 2016). Then, the reads were mapped to conserved protein targets using Diamond v0.9.10 (Buchfink, Xie & Huson, 2014), while paired-end reads were merged when possible by using *bbmerge* from BBtools. The mapped reads were used for subsequent *de novo* assembly for each sample by using SPAdes v3.12.0 (Bankevich et al., 2012); the assemblies were constructed with fixed *kmer* sizes of 25, 35, 45, 55, 65, and 75 bases, while the generated graphs were combined. The selected *de novo* assembly by Diamond software was used as a reference file for mapping trimmed reads with default settings of BWA-MEM (alignment via Burrows-Wheeler transformation) version 0.7.17-r1198-dirty (Li, 2013). This resulted in the complete assembly of the exome (BAM files), where the nucleotide variations (as VCF files) were inferred by using GATK v3.8-1-0 (McKenna et al., 2010) with a minimum base quality of 21. Then, haplotypes and diplotypes were reconstituted from VCF file with GATK and BCFtools v1.7 (Li et al., 2009). After that, all assembled contigs were matched against the protein targets by using Exonerate v2.4.0 (Slater & Birney, 2005), while the best matching contigs with at least 300 nucleotides of perfect matches were mapped against the complete protein set by using Diamond (Buchfink, Xie & Huson, 2014) to identify orthologous sequences. All low-coverage (coverage <10x) nucleotides were replaced with *N*. This exon masking step is required to ensure high data quality and for the efficient multiple sequence alignments in further phylogenetic analysis.

The read depth and heterozygosity for each sample were assessed by using GATK v3.8-1-0 (*CallableLoci*) (McKenna et al., 2010) and Real Time Genomics (RTG) Tools (Cleary et al., 2015) based on the generated BAM and VCF files, respectively. Excess heterozygosity (relative to other samples

of the same species) filters out low-quality assemblies as might arise from sample mixing or poor SNP calling.

3.1.3. Phylogeographic structure and population genetics analysis

Sequence data for the ND4 gene of mtDNA was provided following PCR and Sanger sequencing as described in Potter et al. (2019). The Maximum-Likelihood (ML) phylogenetic tree was constructed using IQTree v1.6.11 (Nguyen, Schmidt, von Haeseler & Minh, 2014) with 1,000 ultrafast bootstraps (Minh et al., 2013) and GTR+G+I model from ModelFinder (Kalyaanamoorthy, Minh, Wong, von Haeseler & Jermini, 2017).

For the nDNA data, the generated haplotype alignments of the exonic regions were filtered by using EAPhy v.1.2 (Blom, 2015), incorporating the alignment by using Muscle v3.8.31 (Edgar, 2004) checking of correct amino acid coding, removal of missing data, and concatenation of sequences. We used the following criteria: maximum missing individuals of 10% and minimum locus length of 200bp, ensuring that each locus is informative for further analysis. The filtered sequences were further cross-checked by eye using Geneious Prime 2020.1.1 (<https://www.geneious.com>) to ensure that the alignments maintained an open reading frame (ORF) with no premature stop codon in the middle of the alignment. Species-level population genetics statistics were inferred based on the haplotypes by using PopGenome (Pfeifer, Wittelsb rger, Ramos-Onsins & Lercher, 2014). The corresponding statistics for northern *Morethia* species were extracted from Potter et al. (2019), which applied very similar informatic and analytical pipelines to exon capture data generated using the same capture probes and sequencing and bioinformatics methods.

Next, the filtered haplotypes for the new southern *Morethia* species were used to infer the diploidy sequence of each sample, using IUPAC ambiguity codes at heterozygous sites (typically <<1% of sites per individual). The concatenated diploidy sequences were then used to construct a ML phylogenetic tree of the southern *Morethia* species by using IQTree v1.6.11 (Nguyen, Schmidt, von Haeseler & Minh, 2014) with 1,000 ultrafast bootstraps (Minh et al., 2013), GTR model, and codon

1+2+3 data partitioning scheme. Then, diplotype sequences were inputted back to EAPhy to extract the list of single nucleotide polymorphism (SNPs) for population genetic analyses. To ensure statistical independence, one SNP was extracted randomly from each locus and then used to infer population structure (i.e. number of lineages within species) based on principal component analysis by using *phangorn* (Schliep, 2010) and *bios2mds* (Pelé, Bécu, Abdi & Chabbert, 2012), as well as model-based clustering using *Structure* (Pritchard, Stephens & Donnelly, 2000) and *conStruct* (Bradburd, Coop & Ralph, 2018). *Structure* analysis incorporated 100,000 burnin steps followed by 1,000,000 MCMC steps with 10 repeats for each value of *K*, ranging from 1 to 6. The best *K* value was extracted via CLUMPAK Server (Kopelman, Mayzel, Jakobsson, Rosenberg & Mayrose, 2015).

In general, there are two major forces that affect population genetic differentiation: isolation by distance (IBD) and local adaptation (Lee & Mitchell-Olds, 2011). In IBD, interbreeding between populations is limited by spatial distance (Wright, 1943): the greater the geographical distance between populations, the less genetic flow between them. It means that two populations that inhabit distant habitats might diverge into two different lineages due to lack of gene flow. As migration is a continuous process across certain landscapes, IBD-derived population differentiation will increase with the increasing distance of the landscape. On the other hand, different habitats might have different ecological and climatic conditions, inducing selection pressure against immigrants (Nosil, Vines & Funk, 2005). As the immigrant (and hybrids) will become less fit due to the natural selection, the opportunities for interbreeding will be limited. This may not relate to physical distance, but rather the combination of abrupt changes that might happen within local environment due to barriers to gene flow and the dispersal behavior of the species, introducing discrete clustering between populations that promotes the maintenance of genetic diversity within population. Based on this idea, it is important to not only cluster the population based on discrete criteria, but also the continuous one (Bradburd, Coop & Ralph, 2018). In *Structure*, the null hypothesis assumes random-mating population (i.e. Hardy-Weinberg equilibrium), which deals with the discrete criteria. On the other hand, *conStruct* is more appropriate for low dispersal, widespread

species as it assumes IBD as the null hypothesis. In order to prevent over-estimation of lineages for species with $K > 1$ in *Structure* outputs, validation via *conStruct* spatial analysis was done, consisting of 5 chains of 100,000 iterations each (20,000 warmup steps) with *adapt_delta* set to 0.999. All visualizations were done in RStudio (RStudio Team, 2015) using the following packages: *fields* (Nychka, Furrer, Paige & Sain, 2017), *rworldmap* (South, 2011), *raster* (Hijmans & van Etten, 2012), and *bayesplot* (Gabry & Mahr, 2019).

3.1.4. Population size analysis

In order to compare the timing of population size changes, Extended Bayesian Skyline Plot (EBSP) of those *Morethia* species with evidence of population expansion (i.e. significant departures from mutation-drift equilibrium via Tajima's D) was constructed based on haplotype sequence data for the 50 longest loci by using BEAST v.2.6.0 (Bouckaert et al., 2014). The analysis was done under HKY model by linking the clock model and unlinking the other parameters, with presumed mutation rate of 0.001 per million years. The chain length was set to be 750,000,000 for each analysis. The results were assessed using Tracer v1.7.1 (Rambaut, Drummond, Xie, Baele & Suchard, 2018) and visualized in RStudio (RStudio Team, 2015).

3.2. Molecular evolution of Australian Eugongylinae skinks

The following analysis utilizes similar approaches to Bromham et al. (2015). The exon-capture sequence alignment of Eugongylinae skinks from Bragg et al. (2018) underwent several filtering steps before used to construct the phylogenetic tree. Then, species pairs were picked based on the tree, followed by calculation of absolute dN and dS, as well as Poisson regression analysis to determine significant predictors of the substitution rates. The complete pipeline is shown in Figure 5.

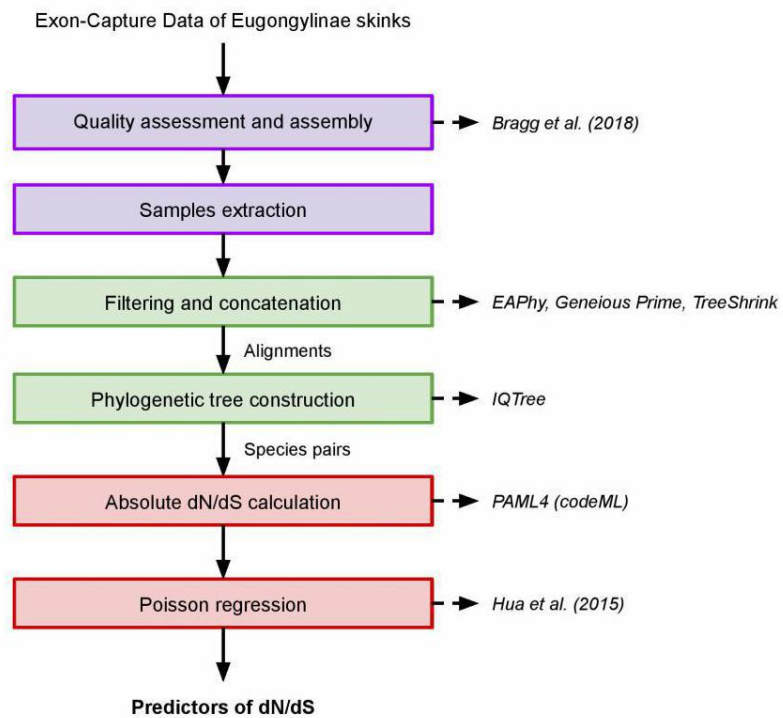


Figure 5. Pipeline of the molecular evolution analysis. Purple: data set; green: data preparation and filtering; red: molecular evolution analysis. The complete lists of software that were utilized can be found in the paragraph below.

3.2.1. Data set

Exon-capture sequences from 21 genera of Eugongylinae were quality-assessed and assembled based on Bragg et al. (2018). The data consists of 443 Australian samples with less than 1000 (out of 3171) missing loci.

3.2.2. Quality assessment and phylogenetics tree construction

The aligned haplotypes (one per individual; h0) of the samples were filtered by using EAPhy v.1.2 (Blom, 2015), incorporating the checking of correct amino acid coding and removal of missing data as above. We used the following criteria: maximum missing individuals of 20% and minimum locus length of 200bp, ensuring that each locus is informative for further analysis. The filtered sequences were further cross-checked by eye using Geneious Prime 2020.1.1 (<https://www.geneious.com>) to ensure correct reading frame and that there is no stop codon in the middle of the alignment.

Then, the gene tree of each locus was constructed by using IQTree v1.6.11 (Nguyen, Schmidt, von Haeseler & Minh, 2014) with MG codon model with additional transition/transversion rate ratio based on the result from ModelFinder (Kalyaanamoorthy, Minh, Wong, von Haeseler & Jermin, 2017). The gene trees and alignments were then inputted into TreeShrink (Mai & Mirarab, 2018) to filter out species with abnormally long branches in each locus. Next, the alignments were inputted into EAPHy to concatenate the sequences, which were then used to construct species tree by using IQTree with GTR model, 1000 ultrafast bootstrap (Minh et al., 2013), and 1+2+3 codon partitioning model.

3.2.3. Choosing species pairs and statistical analysis

Based on the initial tree, we selected species pairs (and a close outgroup) according to these criteria: (i) one well-supported sample was selected to represent one species; (ii) each paired species has distinct trait [i.e. mean annual mean temperature (AMT)] according to the species-level metadata provided to me (Figure 6; Table 1); (iii) species pairs should be as closely-related as possible to reduce node density effects [i.e. underestimation of the number of true changes in long branches due to either sparse sampling or large amount of changes between two successive nodes (Venditti et al., 2006)]; (iv) each species was incorporated in only one pair to ensure independent data points, but can be used as outgroup in others. Mean AMT of each species derived from the distribution records of the species across Australia and indicates a three-fold variation across taxa (Figure 6).

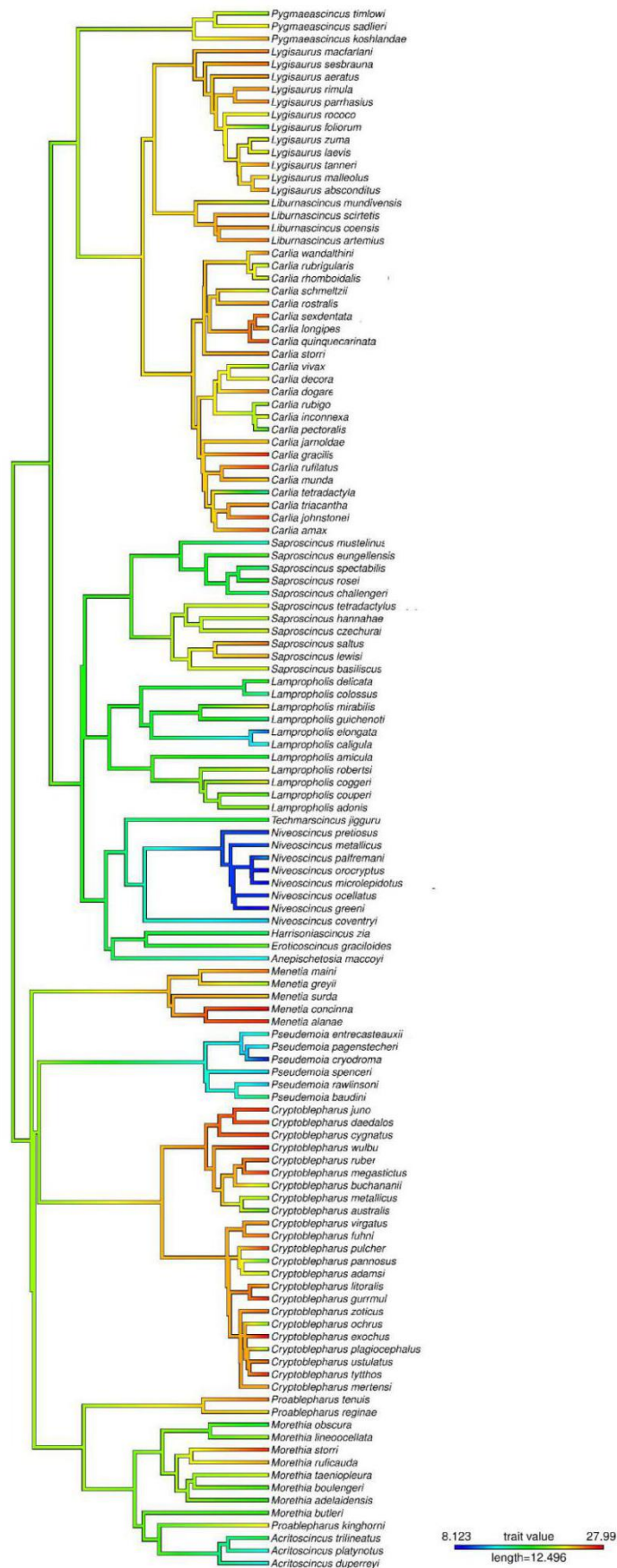


Figure 6. Mean AMT-based species tree of Eugongylinae skinks. Colorization utilizes *phytools* (Revell, 2011).

Table 1. Mean AMT (in degree Celcius) of Eugongylineae skinks.

Genus	Species	Mean AMT	Genus	Species	Mean AMT
<i>Acritosincus</i>	<i>duperreyi</i>	13.998	<i>Liburnascincus</i>	<i>artemius</i>	25.653
	<i>platynota</i>	14.069		<i>coensis</i>	25.056
	<i>trilineata</i>	17.365		<i>mundivensis</i>	22.327
<i>Carlia</i>	<i>amax</i>	26.525		<i>scirtetis</i>	25.233
	<i>decora</i>	23.161	<i>Lygisaurus</i>	<i>absconditus</i>	25.215
	<i>dogare</i>	25.744		<i>aeratus</i>	24.951
	<i>gracilis</i>	27.205		<i>foliorum</i>	19.685
	<i>inconnexa</i>	23.233		<i>laevis</i>	22.631
	<i>jarnoldae</i>	24.485		<i>macfarlani</i>	25.441
	<i>johnstonei</i>	26.816		<i>malleolus</i>	23.512
	<i>longipes</i>	25.176		<i>parrhasius</i>	25.400
	<i>munda</i>	24.289		<i>rimula</i>	25.235
	<i>pectoralis</i>	19.839		<i>rococo</i>	23.184
	<i>quinquecarinata</i>	26.417		<i>sesbrauna</i>	25.790
	<i>rhomboidalis</i>	22.576		<i>tanneri</i>	25.355
	<i>rostralis</i>	25.115		<i>zuma</i>	22.693
	<i>rubigo</i>	21.718	<i>Menetia</i>	<i>alanae</i>	27.261
	<i>rubrigularis</i>	22.432		<i>concinna</i>	27.885
	<i>rufilatus</i>	27.056		<i>greyii</i>	21.774
	<i>schmeltzii</i>	22.337		<i>maini</i>	25.748
	<i>sexdentata</i>	26.337		<i>surda</i>	24.162
	<i>storri</i>	25.109	<i>Morethia</i>	<i>adelaidensis</i>	19.188
	<i>tetradactyla</i>	15.513		<i>boulengeri</i>	19.667
	<i>triacantha</i>	25.382		<i>butleri</i>	19.247
	<i>vivax</i>	21.388		<i>lineoocellata</i>	21.293
	<i>wandalthini</i>	25.477		<i>obscura</i>	17.837
<i>Cryptoblepharus</i>	<i>australis</i>	20.388		<i>ruficauda</i>	24.754
	<i>buchananii</i>	22.163		<i>storri</i>	27.051
	<i>cygnatus</i>	27.083		<i>taeniopleura</i>	22.147
	<i>daedalos</i>	26.874	<i>Niveoscincus</i>	<i>coventryi</i>	12.909
	<i>exochus</i>	27.990		<i>greeni</i>	8.123
	<i>fuhni</i>	25.800		<i>metallicus</i>	10.854
	<i>gurrmul</i>	27.422		<i>microlepidotus</i>	9.039
	<i>juno</i>	27.319		<i>ocellatus</i>	10.009
	<i>litoralis</i>	24.802		<i>orocryptus</i>	8.867
	<i>megastictus</i>	27.106		<i>palfremani</i>	11.850
	<i>mertensi</i>	24.769		<i>pretiosus</i>	10.225
	<i>metallicus</i>	21.748		<i>kinghorni</i>	23.367
	<i>ochrus</i>	21.043	<i>Proablepharus</i>	<i>reginae</i>	23.748
	<i>pannosus</i>	19.996		<i>tenuis</i>	25.837
	<i>plagiocephalus</i>	22.221	<i>Pseudemoia</i>	<i>baudini</i>	16.756
	<i>pulcher</i>	26.891		<i>cryodroma</i>	9.826
	<i>ruber</i>	26.444		<i>entrecasteauxii</i>	14.782
	<i>tytthos</i>	27.096		<i>pagenstecheri</i>	13.948
	<i>ustulatus</i>	25.813		<i>rawlinsoni</i>	12.631
	<i>virgatus</i>	24.973		<i>spenceri</i>	13.271
	<i>wulbu</i>	27.667	<i>Pygmaeascincus</i>	<i>koshlandae</i>	24.822
	<i>zoticus</i>	25.913		<i>sadlieri</i>	23.400
				<i>timlowi</i>	21.212
<i>Erotoscincus</i>	<i>graciloides</i>	20.126	<i>Saproscincus</i>	<i>basiliscus</i>	22.458
<i>Harrisoniascincus</i>	<i>zia</i>	16.916		<i>challengeri</i>	16.733
<i>Lampropholis</i>	<i>adonis</i>	20.905		<i>czechurai</i>	22.338
	<i>amicula</i>	17.646			

	<i>caligula</i>	14.466		<i>eungellensis</i>	20.000
	<i>coggeri</i>	22.989		<i>hannahae</i>	21.788
	<i>colossus</i>	15.600		<i>lewisi</i>	24.595
	<i>couperi</i>	20.156		<i>mustelinus</i>	14.583
	<i>delicata</i>	18.784		<i>rosei</i>	18.776
	<i>elongata</i>	11.233		<i>saltus</i>	25.477
	<i>guichenoti</i>	16.486		<i>spectabilis</i>	16.334
	<i>mirabilis</i>	23.553		<i>tetradactylus</i>	22.596
	<i>robertsi</i>	22.136	<i>Techmarscincus</i>	<i>jigguru</i>	18.400

The absolute number of non-synonymous (dN) and synonymous (dS) substitutions along branches for each species pair were calculated using codeML (Yang, 2007) based on the branch model (i.e. dN/dS on each branch is an independent parameter) using concatenated sequence. Then, Poisson regression analysis based on Hua et al. (2015) was conducted to estimate the association between the absolute dN and dS (response) with mean AMT and other traits [body size, mean nDNA heterozygosity, niche breadth, mean altitude, geographic range, centroid latitude, habit (arboreal, grassy, ground dwelling, leaf litter, saxicola), habitat (coastal, forest, generalist, grassland, rock outcrops, wetland), and biome (arid zone, temperate, tropical mesic, tropical savanna)]. The Poisson regression models substitution as a Poisson process over time, with non-synonymous and synonymous substitution rates of each species in a species pair as a linear function of the species trait on log scale. This allows us to do the analysis without having to estimate the exact divergence time which is unknown (Hua et al., 2015). The statistical significance of each variable is tested by a likelihood ratio test and the biological significance of each variable is estimated by pseudo R^2 , approximating the amount of variation in dN and dS that can be explained by the variable. Lastly, to test if the significance of a variable is due to its covariation with another significant variable (e.g. mean AMT), we performed a likelihood ratio test between a model including both variables and a model only including the variable not under test. A significant result of the likelihood ratio test suggests that the variable under test has statistically significant association with dN and dS that is not due to the covariation between the two variables. We also calculated the difference in the pseudo R^2 between the two models to approximate the additional amount of variation in dN and dS

that can be explained by the variable under test and is not due to the covariation between the two variables.

Chapter IV. Results

4.1. Population structure and phylogenetics of southern *Morethia* skinks

In this part, there were four files produced by the OMG and EAPhy pipelines that were utilized for further analysis: BAM, VCF, diplotypes, and haplotype sequences. The first two were used to count the number of callable loci and within-individual heterozygosity, respectively, while the diplotype/haplotypes were used to infer phylogeographic structure and population dynamics.

4.1.1. Phylogeographic structure

The ND4 mtDNA tree (Figure 7; left) shows that *M. obscura* has the shallowest branches between samples with very low branch supports, inferring very weak phylogeographic structure. This also applies for *M. adalaidensis*, where divergence between blue- and red-colored lineages is weakly supported. On the other hand, *M. butleri* and *M. lineoocellata* consist of two deeply-rooted, well-supported lineages. *M. boulengeri* also has well-defined phylogeographic structure with six lineages; however, several divergence events (e.g. between black- and yellow-colored lineages) have low support values. Nevertheless, this gives an initial idea about the population structure of those species.

From the exon capture data, we extracted 1454 exonic loci after stringent filtering. The estimated exon tree based on the concatenated exons (Figure 7; right) shows well-supported phylogenetic relationship between the species: the first divergence event of *Morethia* happened between *M. butleri* and the most recent common ancestor (MRCA) of the other species. Then, the MRCA sequentially diverged into two major groups. The first diverges into *M. adalaidensis*, and subsequently *M. boulengeri* and *M. taeniopleura*, while the other group has *M. lineoocellata* and *M. obscura* as sister species. This is consistent with the analysis of Potter et al. (2019), where the sister taxa *M. ruficauda* and *M. storri* also diverge from the same MRCA as *M. adalaidensis*. The exon tree only differs with the mtDNA tree at nodes that are weakly supported (e.g. phylogenetic relationship

between *M. adalaidensis*, *M. boulengeri*, and *M. taeniopleura* is different between the trees, but weakly supported in the mtDNA tree). Both trees yield consistent phylogeographic structuring within species: *M. boulengeri*, *M. lineoocellata*, and *M. butleri* have deep, strongly-supported lineages matching the major mtDNA clades; *M. adalaidensis* has intermediate depth and congruence, while *M. obscura* has the shallowest branching and the most incongruent structuring between the trees.

Phylogeographic structure of each species was further assessed by using PCoA and model-based clustering (Figure 8) based on 1454 SNPs (i.e. one SNP per loci). In accordance with the exon tree, *M. obscura* and *M. adalaidensis* have no phylogeographic structuring; all of the samples form one big cluster with no distinctive separation. On the other hand, *M. boulengeri*, *M. butleri*, and *M. lineoocellata* consist of two to three lineages. For *M. boulengeri*, there are three major clusters: samples in South Australia (pink- and purple-colored), samples from western Qld (green-colored), and samples from eastern Australia (black- and red-colored). One sample from northeast South Australia (yellow-colored in Figure 7) groups with eastern samples for mtDNA but appears admixed between the three lineages for nDNA. For *M. butleri*, the samples are separated into the ones in Western Australia (purple-colored) and South Australia (red-colored), while the two clusters in *M. lineoocellata* comprised of the ones in south-west (orange-colored) and mid-west of Western Australia (purple-colored). The samples for *M. taeniopleura* are insufficient for inferring population structure.

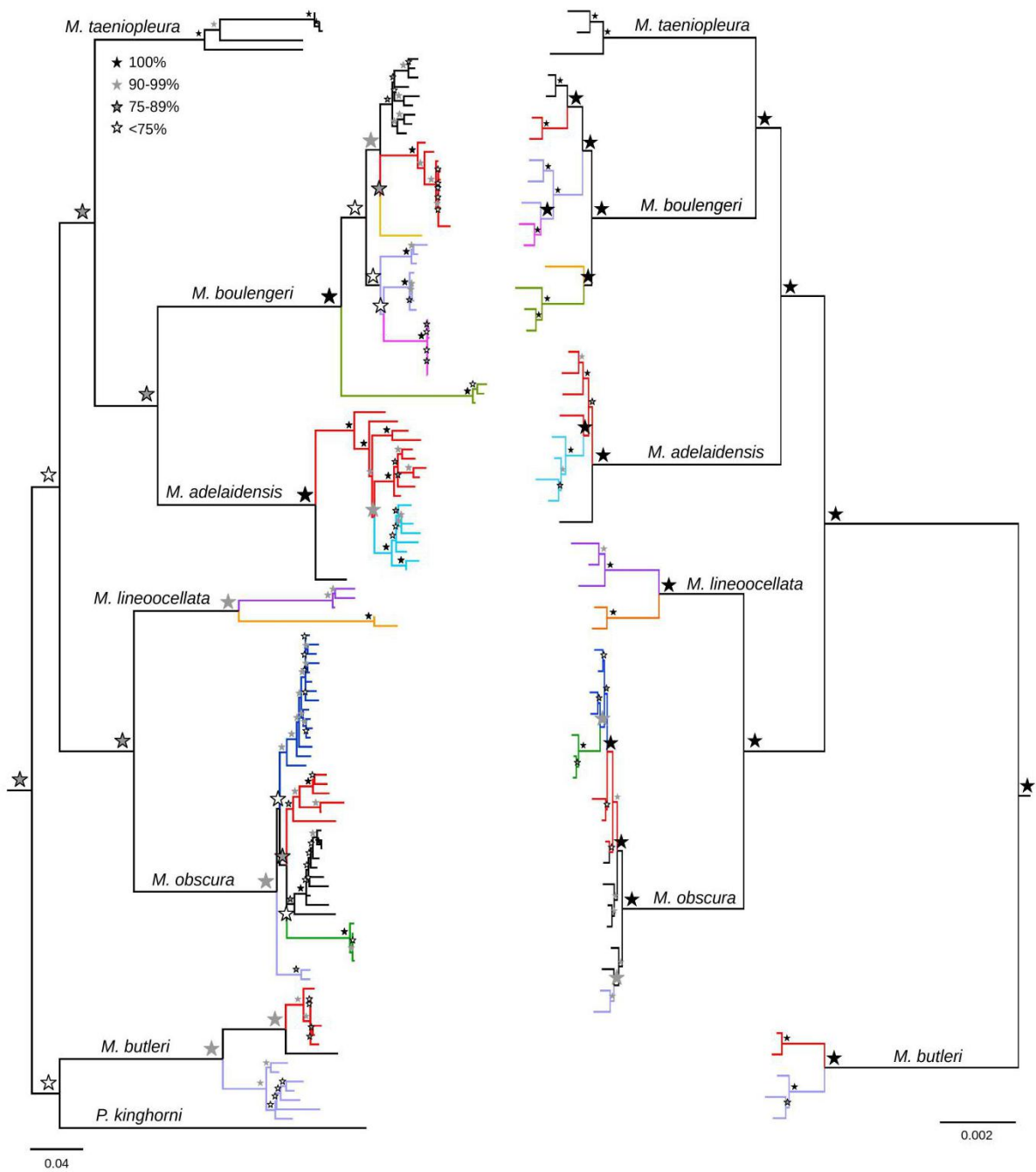


Figure 7. Maximum-likelihood phylogenetic tree of mtDNA (left) and nDNA (right) of southern *Morethia* species. Coloring was based on the phylogeographic clusters of mtDNA. The tree was visualized by using FigTree v.1.4.4 (Rambaut, 2012).

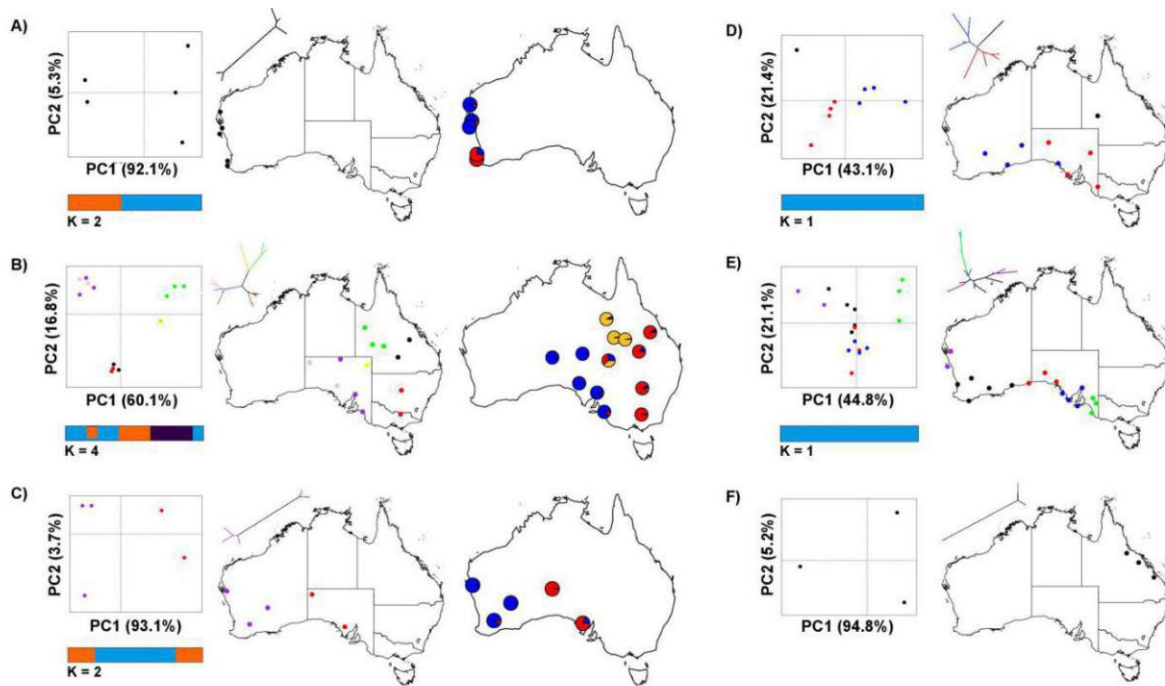


Figure 8. Population structure of southern *Morethia* species. A: *M. lineocellata*, B: *M. boulengeri*, C: *M. butleri*, D: *M. adelaidensis*, E: *M. obscura*, F: *M. taeniopleura*. For A, B, and C, the left map shows the sampling distribution, while the right differentiates lineages based on conStruct analysis.

4.1.2. Population diversity

The within-individual heterozygosity was calculated based on ~2,000,000 nucleotides, with average sequencing depth per sample of 266.75 (115.92 - 434.51) (Table 2). Given this high coverage, we can have considerable confidence in SNP and genotype calls. The average per-individual heterozygosity [(O)Het] of southern *Morethia* species is 0.0030, while the average per-individual heterozygosity by species ranges from 0.0025 (*M. obscura*) to 0.0043 (*M. taeniopleura*; Table 3).

Table 2. Summary statistics of each southern *Morethia* sample based on nDNA data.

Sample ID	Sample Name	Depth	Het. Sites	Callable Loci	O(Het)
SP20_indexing1_S1	A003004_taeniopleura	307.03	9780	2061832	0.47%
SP20_indexing2_S2	ABTC76394_taeniopleura	287.31	4733	2080078	0.23%
SP20_indexing3_S3	ABTC76962_taeniopleura	115.92	13985	2325379	0.60%
SP20_indexing4_S4	WAMR134749_butleri	227.96	7985	1904592	0.42%
SP20_indexing5_S5	WAMR116855_butleri	318.24	4867	1992277	0.24%
SP20_indexing7_S7	ABTC54839_butleri	338.11	6630	2047575	0.32%
SP20_indexing9_S9	ABTC42013_butleri	165.86	10111	1877730	0.54%
SP20_indexing10_S10	ABTC95988_butleri	172.75	4589	1992703	0.23%
SP20_indexing12_S12	ABTC9060_adelaidensis	191.49	5400	2257273	0.24%
SP20_indexing13_S13	ABTC36793_adelaidensis	296.87	8354	2107349	0.40%
SP20_indexing14_S14	ABTC36211_adelaidensis	314.88	6345	2036559	0.31%

SP20_indexing15_S15	ABTC93619_adelaidensis	181.46	7422	2130853	0.35%
SP20_indexing16_S16	ABTC57093_adelaidensis	325.87	5795	1952566	0.30%
SP20_indexing17_S17	WAMR77730_adelaidensis	381.62	6845	2118160	0.32%
SP20_indexing18_S18	WAMR126567_adelaidensis	257.74	3790	2101996	0.18%
SP20_indexing19_S19	WAMR172263_adelaidensis	212.00	6425	2139795	0.30%
SP20_indexing20_S20	ABTC94124_adelaidensis	193.54	6411	2085899	0.31%
SP20_indexing21_S21	WAMR90097_lineocellata	302.85	6872	2061102	0.33%
SP20_indexing22_S22	WAMR140726_lineocellata	201.34	6999	2247966	0.31%
SP20_indexing23_S23	WAMR121093_lineocellata	275.27	8944	1833117	0.49%
SP20_indexing24_S24	WAMR123757_lineocellata	319.74	8997	1917702	0.47%
SP20_indexing25_S25	WAMR127130_lineocellata	300.84	6484	2028877	0.32%
SP20_indexing26_S26	WAMR123044_obscura	297.13	6763	2193369	0.31%
SP20_indexing27_S27	WAMR163182_obscura	238.77	4802	2116632	0.23%
SP20_indexing28_S28	WAMR135144_obscura	250.46	5600	2216943	0.25%
SP20_indexing29_S29	WAMR136392_obscura	244.47	5308	2180366	0.24%
SP20_indexing30_S30	WAMR134127_obscura	347.58	6889	2083917	0.33%
SP20_indexing31_S31	WAMR163633_obscura	316.27	5569	1955999	0.28%
SP20_indexing32_S32	WAMR137788_obscura	213.80	6405	2256824	0.28%
SP20_indexing33_S33	WAMR77874_obscura	218.61	5553	2021398	0.27%
SP20_indexing34_S34	ABTC87391_obscura	300.68	5605	2253153	0.25%
SP20_indexing35_S35	ABTC35117_obscura	179.59	5666	2135140	0.27%
SP20_indexing36_S36	ABTC54848_obscura	232.30	5874	2267376	0.26%
SP20_indexing37_S37	ABTC93654_obscura	196.20	6256	2251127	0.28%
SP20_indexing38_S38	ABTC70381_obscura	241.71	5372	2187254	0.25%
SP20_indexing39_S39	ABTC16707_obscura	324.85	4809	2067326	0.23%
SP20_indexing40_S40	ABTC54805_obscura	225.73	3625	2356627	0.15%
SP20_indexing41_S41	ABTC34455_obscura	261.30	3657	1971275	0.19%
SP20_indexing42_S42	ABTC34479_obscura	329.80	3957	2044568	0.19%
SP20_indexing43_S43	WAMR172297_obscura	238.46	6524	2197523	0.30%
SP20_indexing44_S44	ABTC70372_boulengeri	308.48	3983	2050096	0.19%
SP20_indexing45_S45	ABTC786_boulengeri	298.64	4154	2192948	0.19%
SP20_indexing46_S46	ABTC38482_boulengeri	296.34	7714	2294624	0.34%
SP20_indexing47_S47	ABTC42021_boulengeri	284.07	3418	2067731	0.17%
SP20_indexing48_S48	ABTC127943_boulengeri	314.40	4890	2209043	0.22%
SP20_indexing49_S49	ABTC72764_boulengeri	434.51	4383	2083633	0.21%
SP20_indexing50_S50	A001820_boulengeri	245.70	6800	2181382	0.31%
SP20_indexing51_S51	ABTC9093_boulengeri	264.64	7415	2099094	0.35%
SP20_indexing52_S52	A006510_boulengeri	130.43	7971	2198800	0.36%
SP20_indexing53_S53	A007132_boulengeri	209.85	8062	2160807	0.37%
SP20_indexing54_S54	ABTC3694_boulengeri	296.51	6862	2040059	0.34%
SP20_indexing55_S55	ABTC54774_boulengeri	340.24	6110	2110259	0.29%
SP20_indexing56_S56	ABTC80754_boulengeri	367.63	4076	2113286	0.19%

Table 3. Diversity statistics for exons in Australian *Morethia* species.

Species	#Samples	Average O(Het)	Pi (π)	$\theta(N)$	$\theta(S)$	$\theta(N)/\theta(S)$	Tajima's D	p
<i>M. taeniopleura</i>	3	0.0043	0.0038	0.0011	0.0029	0.096	-0.520	0.357
<i>M. butleri</i>	5	0.0035	0.0034	0.0012	0.0024	0.130	-0.487	0.304
<i>M. lineocellata</i>	5	0.0038	0.0039	0.0011	0.0029	0.103	-0.348	0.389
<i>M. boulengeri</i>	13	0.0027	0.0033	0.0010	0.0024	0.119	-1.420	0.061

<i>M. adelaidensis</i>	9	0.0030	0.0026	0.0008	0.0019	0.119	-1.471	0.059
<i>M. obscura</i>	18	0.0025	0.0018	0.0006	0.0013	0.140	-2.034	0.006
<i>M. ruficauda</i> (AMT)	22	0.0037	0.0028	0.0010	0.0019	0.129	-1.938	0.010
<i>M. ruficauda</i> (AZ)	7	0.0028	0.0019	0.0007	0.0013	0.136	-1.498	0.054
<i>M. storri</i> (AMT)	20	0.0027	0.0042	0.0015	0.0027	0.149	-0.969	0.163

*#Samples: number of samples per species; average O(Het): average per-species observed heterozygosity; π : average number of pairwise differences per site; $\theta(N)$: average diversity of non-synonymous segregation sites; $\theta(S)$: average diversity of synonymous segregation site; Tajima's D: average number of pairwise differences per the number of segregating sites; p: significance value of Tajima's D.

Species-wide nucleotide diversity (π) is greater than mean individual heterozygosity [O(Het)] among *M. bouleengeri* and even more so in *M. storri*. This becomes apparent in species with strong phylogeographic structure. As species with strong phylogeographic structure depart increasingly from a random-mating population, the pairwise difference among populations becomes greater. By referring to the model-based clustering (Figure 8), it might also apply for *M. butleri*, where the difference between π and O(Het) is very slight. This is supported by similar average divergence among lineages between southern and northern species (0.3%-0.5%; Table 4). In contrast, species with weak phylogeographic structure can have higher O(Het) compared to the π . This was found in *M. ruficauda*, *M. adelaidensis*, and *M. obscura* (*M. taeniopleura* is not analyzed due to low number of samples).

Table 4. Lineage-level diversity of Australian *Morethia* species.

Species	No. lineages	Overall π	Mean π within lineages	Mean Dxy between lineages	$\theta(N)/\theta(S)$	Lineages with significant Tajima's D
<i>M. taeniopleura</i>	1	0.0038	N/A	N/A	0.096	
<i>M. butleri</i>	2	0.0034	0.0027	0.0041	0.130	
<i>M. lineocellata</i>	2	0.0039	0.0026	0.0048	0.103	
<i>M. bouleengeri</i>	3	0.0033	0.0022	0.0038	0.119	
<i>M. adelaidensis</i>	1	0.0026	N/A	N/A	0.119	Species (-1.471)
<i>M. obscura</i>	1	0.0018	N/A	N/A	0.140	Species (-2.034)
<i>M. ruficauda</i> (AMT)	4	0.0028	0.0021	0.0031	0.129	AMT (-1.938)
<i>M. ruficauda</i> (AZ)	1	0.0019	N/A	N/A	0.136	AZ (-1.498)
<i>M. storri</i> (AMT)	4	0.0042	0.0017	0.0049	0.149	

*No. lineages: number of lineages per species; overall π : average number of pairwise differences per site per species; mean π within lineage: average number of pairwise differences per site per lineage; mean Dxy between lineages: genetic distance between lineages; $\theta(N)$: average diversity of non-synonymous segregation sites; $\theta(S)$: average diversity of synonymous segregation site.

The non-synonymous sites in all *Morethia* species have lower diversity than the synonymous ones, with the average $\theta(N)/\theta(S)$ ratio of 0.118 (southern species), 0.138 (northern species), and 0.125 (all). This is as expected, as non-synonymous mutations might be deleterious, and subject to purifying selection. An elevated value can be expected in species that undergo strong population fluctuations, such as drift in small populations and founder event during expansion. However, it is not obvious here. The diversity of non-synonymous mutations also has a wider distribution (variation) across loci than the synonymous ones, inferring variation in strength of the selection that works on each locus. The distribution graph of each species is shown in Figure 9.

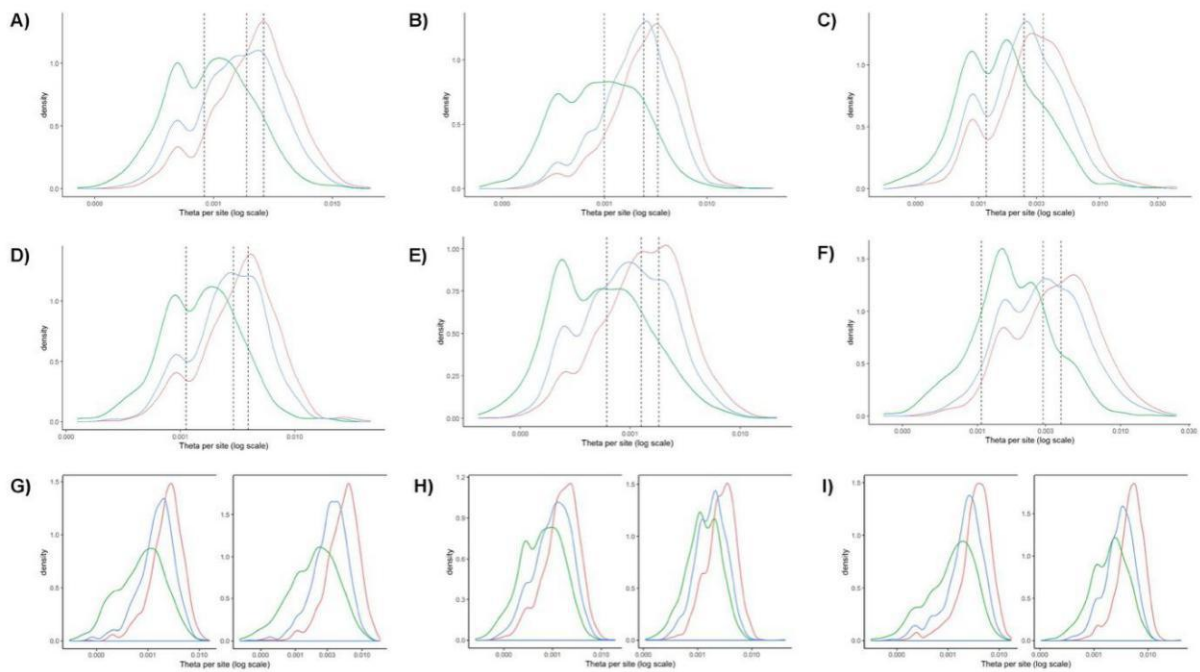


Figure 9. Estimates of diversity (theta) of *Morethia* species. A: *M. adelaidensis*, B: *M. boulengeri*, C: *M. butleri*, D: *M. lineocellata*, E: *M. obscura*, F: *M. taeniopleura*, G: *M. ruficauda* (AMT), H: *M. ruficauda* (AZ), and I: *M. storri* (AMT). For Northern species, the left and right plots showed Tajima's and Watterson's Theta, respectively. Red: all sites; blue: synonymous sites; green: non-synonymous sites.

4.1.3. Population size over time

There are two species that have significantly negative Tajima's D, inferring a strong departure from mutation-drift equilibrium; *M. obscura* and *M. ruficauda* (AMT), whereas *M. adelaidensis* and *M. ruficauda* (AZ) has nearly significant Tajima's D (Table 3). Accordingly, the EBSPs

for *M. adalaidensis* and *M. obscura* (Figure 10A-B) infer recent rapid population growth for both species. *M. adalaidensis* started to expand around 1.25 Mya, while *M. obscura* underwent more recent population expansion (~0.3 Mya). On the other hand, arid zone *M. ruficauda* does not show rapid population growth (Figure 10C). Instead, the expansion started at around 2 Mya (AZ) and has been continuous since then. The selected loci might not be informative enough to reflect recent population expansion events. The AMT lineages of *M. ruficauda* also started rapid growth in total Ne ~2 Mya, with recent population depletion around 250 kya (Figure 10D). More loci should be incorporated in further analysis to better evaluate the results.

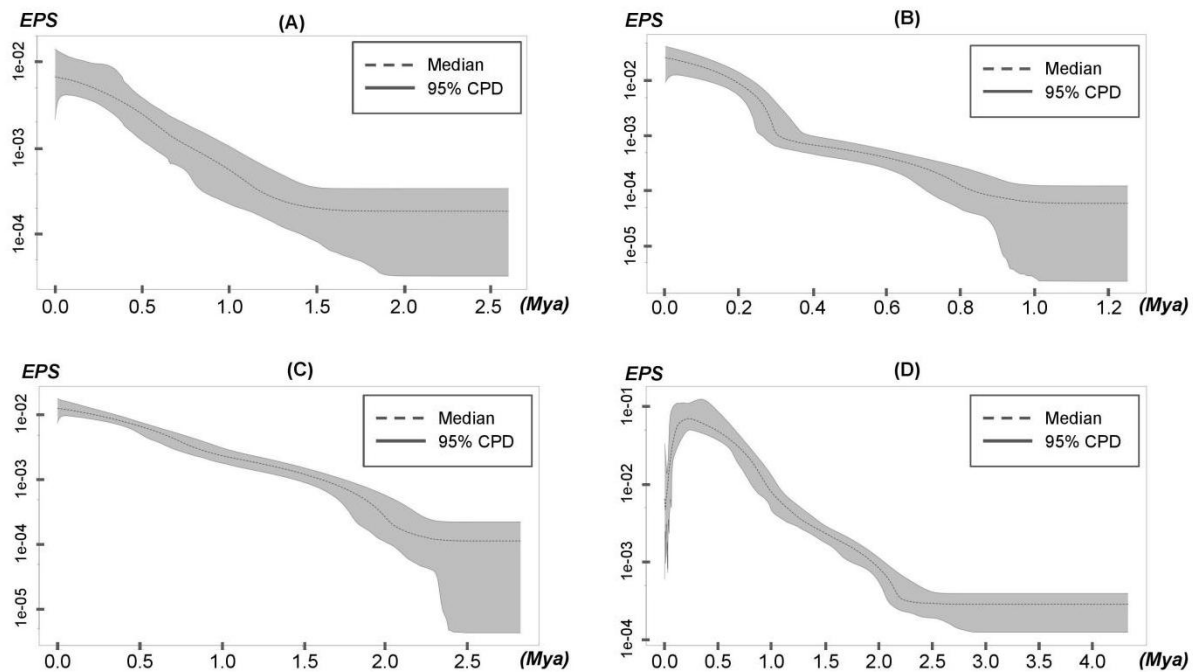


Figure 10. Extended Bayesian Skyline Plot of *M. adalaidensis* (A), *M. obscura* (B), arid zone *M. ruficauda* (C), and monsoonal *M. ruficauda* (D). CPD: central posterior density; EPS: effective population size; Mya: million years ago.

4.2. Molecular evolution of Australian Eugongylinae skinks

After stringent filtering, we extracted 1268 loci and 44 species pairs from the data set. The distribution of temperature contrast between our species pairs and randomly-selected pairs is shown in Figure 11. In general, the data distribution from both groups follows similar trends, peaking at differences of between 0 and 5° Celsius. Randomly-selected species pairs have higher densities

between 10-20° Celsius than our species pairs. However, it does not consider the node density effect as explained before.

The phylogenetic tree of our species pairs and the metadata of each species are shown in Figure 12, with absolute dN and dS in Table 5. As shown in Figure 12, species that live in cold (blue) environment (e.g. *Niveoscincus*) generally have shorter branch lengths than the ones from temperate (e.g. *Saproscincus*) and hot (red) environments (e.g. *Menetia*). This might infer the association between temperature (mean AMT) and substitution rates, where higher temperature induces faster substitution rates, resulting in longer branches. The scatter plot between contrast in dN/dS with contrast in mean AMT (main variable of interest) of each species pair is shown in Figure 13, while the Poisson regression results between absolute dN and dS with predictors (mean AMT, life-history traits) are shown in Table 6-7.

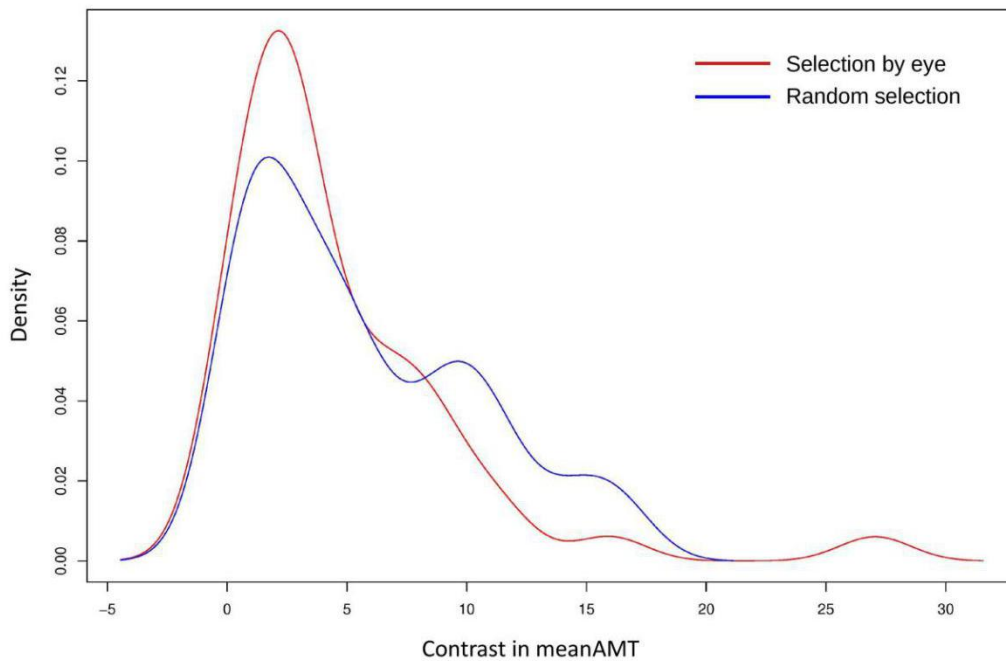


Figure 11. Density plot of mean AMT contrast between selection by eye (our species pairs) and random selection. The bandwidth was 1.5.

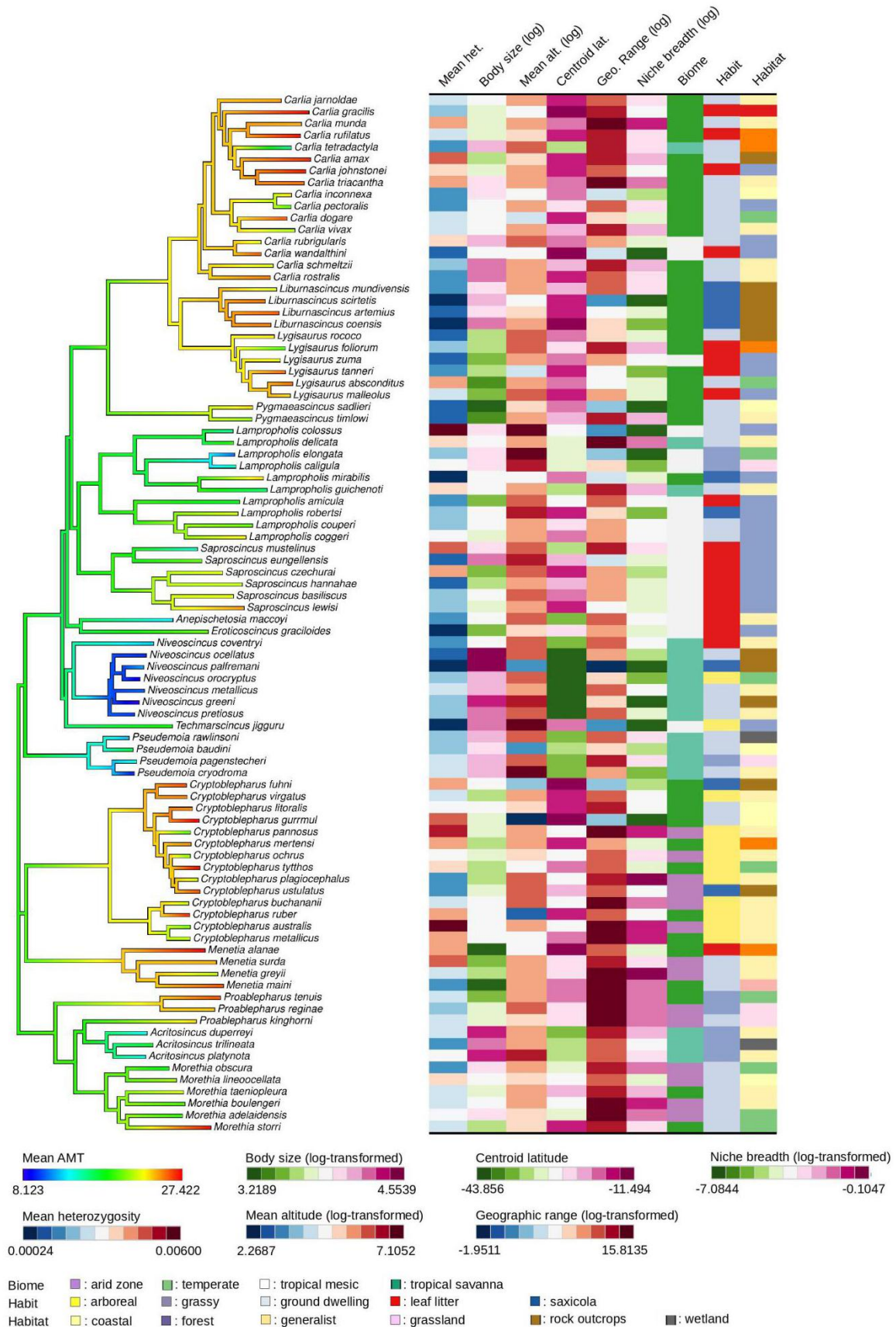


Figure 12. Phylogenetic tree of Eugongyliinae species pairs. The tree is colored based on mean AMT of each species. Unlabeled colors on *habitat* are species that live in multiple habitats and not used in the analysis. Colorization utilizes *phytools* (Revell, 2011) and *RColorBrewer* (Neuwirth, 2014).

Table 5. Absolute dN and dS of each species pair.

Species		Absolute dN		Absolute dS	
A	B	A	B	A	B
<i>Mo. storri</i>	<i>Mo. adelaidensis</i>	1723.7	1139.7	5149.9	3399.4
<i>Mo. boulengeri</i>	<i>Mo. taeniopleura</i>	1187.1	1111.5	3517.4	3513.0
<i>Mo. lineoocellata</i>	<i>Mo. obscura</i>	912.1	852.4	3096.2	2501.3
<i>Ac. trilineata</i>	<i>Ac. platynota</i>	860.3	645.1	2233.1	1825.9
<i>Pr. kinghorni</i>	<i>Ac. duperreyi</i>	2360.1	1415.0	6549.8	3774.5
<i>Pr. reginae</i>	<i>Pr. tenuis</i>	1089.4	1152.7	3244.4	3615.8
<i>Me. maini</i>	<i>Me. greyii</i>	1218.9	1216.3	4102.3	3828.4
<i>Me. alanae</i>	<i>Me. surda</i>	1810.9	1756.7	5182.7	5353.3
<i>Cr. metallicus</i>	<i>Cr. australis</i>	492.1	385.5	1235.0	1264.3
<i>Cr. ruber</i>	<i>Cr. buchananii</i>	422.4	490.3	1638.2	1668.0
<i>Cr. ustulatus</i>	<i>Cr. plagiocephalus</i>	500.6	435.7	1288.5	1232.3
<i>Cr. tythos</i>	<i>Cr. ochrus</i>	496.7	372.0	1602.7	1117.4
<i>Cr. mertensi</i>	<i>Cr. pannosus</i>	540.8	598.8	1966.6	1961.9
<i>Cr. gurrumul</i>	<i>Cr. litoralis</i>	662.5	501.5	1572.9	1184.7
<i>Cr. virgatus</i>	<i>Cr. fuhni</i>	522.4	566.6	1763.5	1891.7
<i>Ps. cryodroma</i>	<i>Ps. pagenstecheri</i>	404.5	515.0	1088.3	1418.9
<i>Ps. baudini</i>	<i>Ps. rawlinsoni</i>	618.6	501.0	1569.0	1386.1
<i>N. greeni</i>	<i>N. metallicus</i>	682.3	891.8	1222.6	1527.1
<i>N. orocryptus</i>	<i>N. palfremani</i>	411.4	515.0	819.1	990.7
<i>N. ocellatus</i>	<i>N. coventryi</i>	1831.8	1985.4	3785.5	4299.1
<i>E. graciloides</i>	<i>An. maccoyi</i>	3110.2	2682.5	7243.9	5145.8
<i>S. lewisi</i>	<i>S. basiliscus</i>	1658.5	1228.1	4498.8	3475.4
<i>S. eungellensis</i>	<i>S. mustelinus</i>	1738.0	1645.9	3768.1	3616.5
<i>La. coggeri</i>	<i>La. couperi</i>	1180.5	1379.4	3673.1	4195.1
<i>La. robertsi</i>	<i>La. amacula</i>	2196.7	2275.5	6145.9	6258.2
<i>La. guichenoti</i>	<i>La. mirabilis</i>	1391.4	1250.2	4125.2	4001.7
<i>La. caligula</i>	<i>La. elongata</i>	512.2	526.8	1431.5	1361.2
<i>La. delicata</i>	<i>La. colossus</i>	480.9	508.0	1861.0	1817.1
<i>Py. timlowi</i>	<i>Py. sadlieri</i>	816.9	880.2	2506.2	2494.6
<i>Ly. malleolus</i>	<i>Ly. absconditus</i>	365.9	377.9	1318.2	1370.4
<i>Ly. zuma</i>	<i>Ly. tanneri</i>	694.2	728.2	2142.0	2444.7
<i>Ly. foliorum</i>	<i>Ly. rococo</i>	873.6	881.9	2823.3	2760.2
<i>Li. coensis</i>	<i>Li. artemius</i>	778.0	887.5	2358.1	2644.5
<i>Li. mundivensis</i>	<i>Li. scirtetis</i>	1020.2	855.8	3416.0	2419.7
<i>Ca. rostralis</i>	<i>Ca. schmeltzii</i>	1034.1	1224.5	3810.1	3990.8
<i>Ca. wandalthini</i>	<i>Ca. rubrigularis</i>	449.4	541.4	1585.7	1512.2
<i>Ca. vivax</i>	<i>Ca. dogare</i>	1078.5	870.1	3545.5	2975.7
<i>Ca. pectoralis</i>	<i>Ca. inconnexa</i>	354.6	308.1	909.4	847.1
<i>Ca. amax</i>	<i>Ca. tetradactyla</i>	1487.6	1108.0	4446.6	3445.5
<i>Ca. rufilatus</i>	<i>Ca. munda</i>	938.6	979.1	3310.5	3396.4

<i>Ca. triacantha</i>	<i>Ca. johnstonei</i>	895.0	907.0	2916.7	3006.0
<i>Ca. jarnoldae</i>	<i>Ca. gracilis</i>	1082.2	1691.4	3803.5	5267.2
<i>N. pretiosus</i>	<i>T. jigguru</i>	1677.4	2268.0	3425.6	6152.4
<i>S. hannahae</i>	<i>S. czechurai</i>	1511.6	1271.3	4388.8	3218.1

*Ac: *Acritosincus*; An: *Anepischetosia*; Ca: *Carlia*; Cr: *Cryptoblepharus*; E: *Erotoscincus*; La: *Lampropholis*; Li: *Liburnascincus*; Ly: *Lygisaurus*; Me: *Menetia*; Mo: *Morethia*; N: *Niveoscincus*; Pr: *Proablepharus*; Ps: *Pseudemoia*; Py: *Pygmaeascincus*; S: *Saproscincus*; T: *Techmarscincus*

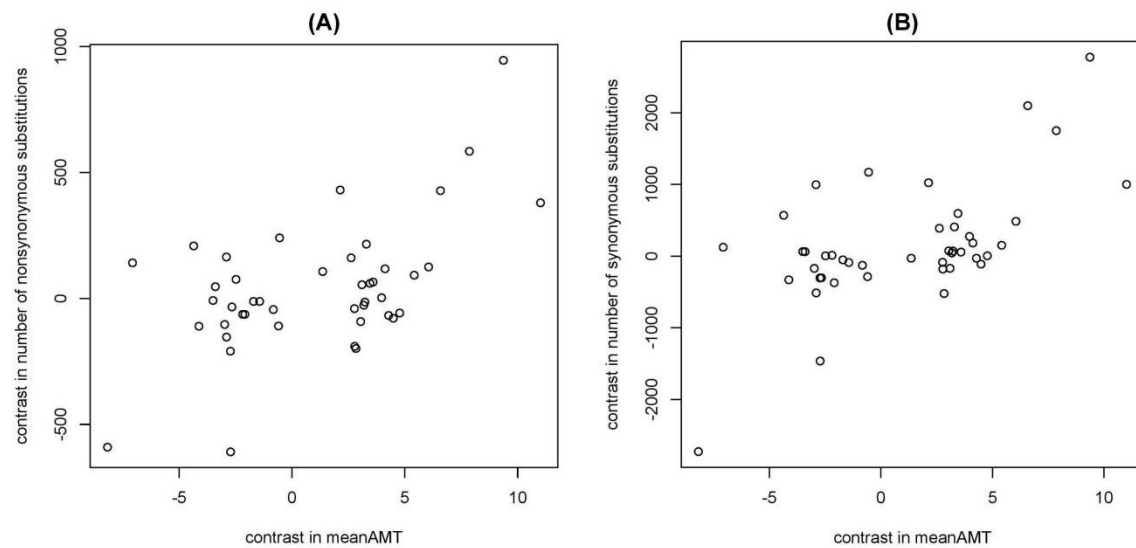


Figure 13. Scatter plot of contrasts across independent species pairs between mean AMT with dN (A) and dS (B) of the species pairs.

Table 6. Poisson regression results of mean AMT and life-history traits on absolute dN/dS.

Predictors	Response	Mean	SE	LR	p	R ²
Mean AMT	dN	0.0518	0.0027	366.8343	<0.001	0.3540
	dS	0.0658	0.0016	1630.3690	<0.001	0.4439
Mean heterozygosity ^a	dN	-0.0247	0.0089	7.6421	0.0057	0.0074
	dS	-0.0477	0.0054	78.6738	<0.001	0.0214
Body size ^b	dN	-0.8791	0.0492	320.7286	<0.001	0.3095
	dS	-1.0902	0.0305	1289.5900	<0.001	0.3511
Geographic range ^b	dN	-0.0121	0.0028	18.2370	<0.001	0.0176
	dS	-0.0175	0.0017	106.1221	<0.001	0.0289
Centroid latitude	dN	0.0209	0.0014	235.7218	<0.001	0.2275
	dS	0.0289	0.0008	641.1501	<0.001	0.1746
Mean altitude ^b	dN	-0.1189	0.0141	71.0332	<0.001	0.0685
	dS	-0.1048	0.0085	152.7191	<0.001	0.0416
Niche breadth ^b	dN	-0.0059	0.0070	0.7143	0.3980	0.0007
	dS	-0.0218	0.0041	28.4621	<0.001	0.0077
Biome	dN	See Table 7-9		238.2640	<0.001	0.2299
	dS			1058.8400	<0.001	0.2883
Habit	dN			208.1081	<0.001	0.2008

	dS		964.8584	<0.001	0.2627
	dN		210.5153	<0.001	0.2031
Habitat	dS		835.9470	<0.001	0.2276

^amean heterozygosity*1000; ^blog transformed; AMT: annual mean temperature; Mean: mean of the regression coefficient; SE: standard error of the regression coefficient; LR: likelihood ratio; *p*: p-value of the likelihood ratio test; *R*²: pseudo *R*²

Table 7. Poisson regression results of pairwise biome comparisons on absolute dN (above gray area) and dS (below gray area). Value shows the mean of the regression coefficient of each biome relative to the reference biome. The reference biome is shown as columns. The order of both synonymous and nonsynonymous substitution rates is T<AZ<=TM=TS.

Biomes		dN			
		AZ	T	TM	TS
dS	AZ		0.2493	0.0024	-0.0299
	T	-0.2778		-0.2469	-0.2792
	TM	0.0757	0.3534		-0.0320
	TS	0.0488	0.3266	-0.0269	

*AZ: arid zone; T: temperate; TM: tropical mesic; TS: tropical savanna

Table 8. Poisson regression results of habit on absolute dN (above gray area) and dS (below gray area). Value shows the mean of the regression coefficient of each habit relative to the reference habit. The reference habit is shown as columns. The order of both synonymous and nonsynonymous substitution rates is G<GD<L=S<A.

Habit		dN				
		A	G	GD	L	S
dS	A		0.0394	0.1433	0.0581	0.0649
	G	-0.7219		-0.3594	-0.4446	-0.4377
	GD	-0.3362	0.3860		-0.0852	-0.0783
	L	-0.2461	0.4761	0.0901		0.0069
	S	-0.1945	0.5278	0.1417	0.0517	

*A: arboreal; G: grassy; GD: ground dwelling; L: leaf litter; S: saxicola

Table 9. Poisson regression results of habitat on absolute dN (above gray area) and dS (below gray area). Value shows the mean of the regression coefficient of each habitat relative to the reference habitat. The reference habitat is shown as columns. The order of both synonymous and nonsynonymous substitution rates is G<R<=W=F<=C<GR.

Habitat		dN					
		C	F	G	GR	R	W
dS	C		0.5556	0.1442	-0.1387	0.0699	0.0082
	F	-0.0527		0.0886	-0.1942	0.0144	-0.0473
	G	-0.1807	-0.1279		-0.2829	-0.0742	-0.1359
	GR	0.1253	0.1780	0.3060		0.2088	0.1470
	R	-0.0906	-0.0379	0.0901	-0.2159		-0.0617
	W	-0.0405	0.0122	0.1402	-0.1658	0.0502	

*C: coastal; F: forest; G: generalist; GR: grassland; R: rock outcrops; W: wetland

As shown in Table 6, there are six variables that significantly explain the variation in dN and dS: mean AMT (R^2 : dN 35.40%, dS 44.39%), body size (30.95%, 35.11%), centroid latitude (22.75%, 17.46%), biome (22.99%, 28.83%), habit (20.08%, 26.27%), and habitat (20.31%, 22.76%). Variables that could be related to population size - mean individual heterozygosity, geographic range size, and niche breadth - explained little of the variation in dN (<2%) or dS (<3%).

In general, nonsynonymous and synonymous substitutions share the same variation patterns across species. Species distributed in higher temperature, near the tropics, with smaller body size tend to have higher substitution rates. For biome, species in tropical savanna and tropical mesic zones have the highest substitution rates, while temperate species having the lowest substitution rates (Table 7). For habit, arboreal species have the highest substitution rates, while species in grassy habit having the lowest substitution rates (Table 8). For habitat, grassland species have the highest substitution rates, while generalist having the lowest substitution rates (Table 9).

After differences in temperature (mean AMT) was controlled for statistically, the other significant variables explain less than 10% of the variation in dN and dS (Table 10). In contrast, differences in temperature still explain about 20% variation in dN and dS above and beyond what can be explain by each of the other variables. This infers that temperature is the strongest predictor of both dN and dS in *Eugongylinae* skinks. However, we cannot completely remove other variables from the equation, as their results are still statistically significant after controlling for differences in temperature. In particular, body size, as expected from Bergmann's Rule (Bergmann, 1847), has a significant negative correlation with temperature (Figure 14; coefficient = -0.587, $p < 0.001$), and it has the highest explanatory power of both dN and dS above and beyond its covariation with temperature (~7%), given that it has fewer parameters than the model with habit (Table 10).

After controlling for the difference in temperature, tropical mesic species still have the highest substitution rates, but tropical savanna species no longer have higher substitution rates than temperate species (Table 11). Species in grassy habit still have the lowest substitution rates, but the

rates are similar to species in *rocky* (saxicolous) habit (Table 12). Generalists no longer have the lowest substitution rates, but similar rates as species in other habitats (Table 13).

Table 10. Poisson regression results on testing significance due to covariation between predictors..

Predictors		Response	Mean	SE	LR	<i>p</i>	R ²
#1	Mean AMT	dN	0.0359	0.0033	120.2268	<0.001	0.1160
	Body size ^b		-0.5123	0.0595	74.1184	<0.001	0.0715
	Mean AMT	dS	0.0481	0.0019	586.7610	<0.001	0.1598
	Body size ^b		-0.5808	0.0370	245.9740	<0.001	0.0669
#2	Mean AMT	dN	0.1013	0.0076	180.7383	<0.001	0.1744
	Centroid latitude		-0.0268	0.0038	49.6231	<0.001	0.0479
	Mean AMT	dS	0.0991	0.0046	1050.5919	<0.001	0.2861
	Centroid latitude		-0.0181	0.0023	61.3650	<0.001	0.0167
#3	Mean AMT	dN	0.0501	0.0039	159.9247	<0.001	0.1543
	Biome		See Table 11		31.3517	<0.001	0.0303
	Mean AMT	dS	0.0669	0.0024	767.9820	<0.001	0.2091
	Biome		See Table 11		196.4450	<0.001	0.0535
#4	Mean AMT	dN	0.0518	0.0033	241.0994	<0.001	0.2326
	Habit		See Table 12		82.3705	<0.001	0.0795
	Mean AMT	dS	0.0616	0.0020	951.5756	<0.001	0.2591
	Habit		See Table 12		286.0570	<0.001	0.0779
#5	Mean AMT	dN	0.0467	0.0034	193.5512	<0.001	0.1868
	Habitat		See Table 13		37.2295	<0.001	0.0359
	Mean AMT	dS	0.0611	0.0020	902.4310	<0.001	0.2457
	Habitat		See Table 13		108.0010	<0.001	0.0294

^blog transformed; mean: mean of regression coefficient; SE: standard error of regression coefficient; LR: likelihood ratio of the model against a model only including the other predictor; *p*: p-value of the likelihood ratio test; R²: difference in pseudo R² between the model and a model only including the other predictor

Table 11. Poisson regression results of pairwise biome comparisons on absolute dN (above gray area) and dS (below gray area), after controlling for differences in temperature. Value shows the mean of the regression coefficient of each biome relative to the reference biome. The reference biome is shown as columns. The order of both synonymous and nonsynonymous substitution rates is T=TS<AZ<=TM.

Biomes		dN			
		AZ	T	TM	TS
dS	AZ		0.0767	-0.0128	0.0579
	T	-0.0549		-0.0894	-0.0187
	TM	0.1098	0.1649		0.0707
	TS	-0.0694	-0.0142	-0.1790	

*AZ: arid zone; T: temperate; TM: tropical mesic; TS: tropical savanna

Table 12. Poisson regression results of pairwise habit comparisons on absolute dN (above gray area) and dS (below gray area), after controlling for differences in temperature. Value shows the mean of the regression coefficient of each habit relative to the reference habit. The reference habit is shown as columns. The order of both synonymous and nonsynonymous substitution rates is GR<=S<G<=A=L.

Habit		dN				
		A	G	GD	L	S
dS	A		0.1517	-0.0347	-0.0645	0.0815
	G	-0.3077		-0.1864	-0.2162	-0.0701
	GD	-0.1273	0.1804		-0.0298	0.1162
	L	-0.0990	0.2087	0.0283		0.1461
	S	-0.2120	0.0957	-0.0847	-0.1129	

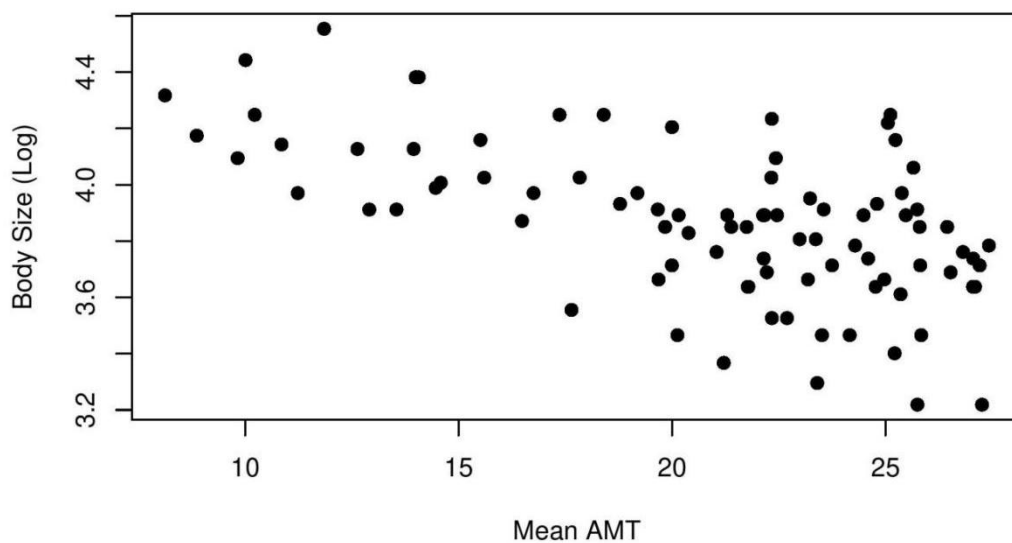
*A: arboreal; G: grassy; GD: ground dwelling; L: leaf litter; S: saxicola

Table 13. Poisson regression results of pairwise habitat comparisons on absolute dN (above gray area) and dS (below gray area), after controlling for differences in temperature. Value shows the mean of the regression coefficient of each habitat relative to the reference habitat. The reference habitat is shown as columns. The order of both synonymous and nonsynonymous substitution rates is R<=G<C<=F<=W<=GR.

Habitat		dN					
		C	F	G	GR	R	W
dS	C		0.0149	0.0182	-0.1195	0.0278	-0.0610
	F	0.0027		0.0033	-0.1344	0.0129	-0.0759
	G	-0.0181	-0.0208		-0.1377	0.0096	-0.0792
	GR	0.1047	0.1021	0.1228		0.1473	0.0585
	R	-0.0665	-0.0691	-0.0484	-0.1710		-0.0888
	W	0.0485	0.0458	0.0666	-0.0560	0.1151	

*C: coastal; F: forest; G: generalist; GR: grassland; R: rock outcrops; W: wetland

Figure 14. Scatter plot of mean AMT and body size of the species pairs.



Chapter V. Discussion

5.1. Population structure and phylogenetics of southern *Morethia* skinks

In comparison with *Morethia* species from the AMT studied by Potter et al. 2019, southern *Morethia* species have moderate population structures with one to three lineages per species. *M. adalaidensis* and *M. obscura* have no population structure, which might infer two possibilities: a recent expansion event from a single refugium or old, large panmictic population. By looking at the tree, the first is more likely, as each species has short branch lengths between samples. This is supported statistically by the low value of π compared to $O(Het)$. On the other hand, *M. boulengeri*, *M. butleri*, and *M. lineocellata* respectively have three, two, and two independent lineages, which are more pronounced than the northern AZ *M. ruficauda*, but weaker than the AMT *M. storri* in which there are multiple microendemic lineages in mesic regions. This is reflected from the clustering models (i.e. *Structure* and *conStruct*) and high nucleotide diversity between populations. The moderate to strong structure of *M. boulengeri* compared to other widespread AZ taxa (*M. adalaidensis*, *M. butleri*, and arid zone *M. ruficauda*) confirms a greater local persistence in eastern AZ, which might be correlated with the presence of lakes and rivers in the eastern Australia through the LGM (Fitzsimmons et al., 2013), providing suitable habitat and allowing the species to inhabit a wider landscape. While further population subdivision might be present within the species (Janes et al., 2017), more sampling is required to better evaluate this problem.

Evidence of recent population expansion is found in three species: *M. obscura*, *M. ruficauda*, and potentially *M. adalaidensis*, denoted by significant negative Tajima's D value. However, the expansion timing between the northern and southern species is different. Southern *M. obscura* underwent more recent population expansion compared to the northern *M. ruficauda*. This might be correlated with the aridification history of the continent (Fitzsimmons et al. 2013). Warmer habitat in the northern area would allow early expansion for the species, followed by continuous population growth since then. In contrary, the renewed aridity for southern Australia in late

Holocene forced the species to stay in refugia, before expanding in more recent years. The evidences of population expansion itself become apparent for *M. ruficauda* that live in the AMT and (northern) AZ, as extreme aridity during the LGM would restrict the species to geographically limited mesic refugia (see also Potter et al., 2018). However, the sudden decrease of effective population size (N_e) of monsoonal *M. ruficauda* could also be due to recent population merging, which results in the decrease of total N_e . In the case of *M. obscura* and *M. adelaidensis*, stronger population fluctuations might be related to the limited climatic niche (*M. obscura*) or habitat preference (*M. adelaidensis*) of these species.

Compared to other species whose distribution ranges from dry to arid zones, *M. obscura* is mainly found in coastal semi-arid regions of southern, eastern, and western parts of Australia (Cogger, 2014). If the restricted spatial distribution also reflects a more specialized physiology of the species, this might limit the adaptive ability of the species to new refugia during extreme climatic conditions. According to Greer (1980), it was indeed the case, where the critical maximum temperature of *M. obscura* (42.3°C) was lower than the other *Morethia* species such as *M. butleri* and *M. lineocellata* (44.6°C). Combined with limited habitat and niche during the climatic extremes, *M. obscura* would likely to contract to single refugium and expand its range when the environmental condition became more favorable. Specifically, samples of *M. obscura* that live in South Australia have lower mean heterozygosity (0.23%) than the ones in western part of Australia (0.28%) (Table 1). As population expansion would result in the decrease of heterozygosity (due to adaptive requirement to the new environment), *M. obscura* might have been restricted in the western part of Australia during the LGM and then expanded to the east during the Holocene. According to Keast (1961) and Ford (1987), the South-West (SW) and Pilbara regions were two possible areas of refugia during the Pleistocene, denoted by high rainfall in the present-day. Of these two areas, *M. obscura* has a high density in the SW refugium, supported by the result of Maxent model analysis (Figure 2E). On the other hand, *M. adelaidensis* is mainly found in the semi-arid to arid area of southern Australia (Cogger, 2014). Even though its distribution is quite vast, habitat of *M. adelaidensis* is

limited to the chenopod shrubland. This habitat restriction might heavily impact the species during the climate fluctuations. One of the potential refugium might be the Nullarbor Plain. While it has been identified as geographical barrier between south-west and south-east area, some species' remains were found in the Nullarbor Plain caves and dolines (Geosence International, 2009), which might suggest its suitable condition for organisms in the past. Alternatively, the species might migrate to the more mesic area around Eyre Peninsula and/or Mount Lofty Ranges during the harsh period (Keast, 1961; Ford, 1987). In comparison with *M. obscura*, *M. adalaidensis* individuals that live in South Australia have higher heterozygosity (0.34%) than the ones living in Western Australia (0.28%). These results might give an idea on the directionality of *M. obscura* and *M. adalaidensis* contraction and expansion. Further analysis should incorporate more samples with explicit coalescent modelling (e.g. Peter & Slatkin (2013) spatial expansion model) to explore areas of refugia across the landscape.

In general, southern *Morethia* species have generally lower $\theta(N)/\theta(S)$ ratio than their northern counterparts. This supports the weak-to-moderate structuring of the species, allowing purifying selection to work more effectively in larger effective population sizes. Interestingly, *M. obscura* and *M. adalaidensis* have the lowest within-species nucleotide diversity among the southern *Morethia* species, but also the highest and third highest $\theta(N)/\theta(S)$ ratio, respectively. This might explain bottleneck event during LGM: when most of the habitat became inhospitable, the population size of the species would be greatly reduced. This would promote genetic drift and thus limiting the effect of purifying selection. During the spatial expansion after LGM, it would start a new colony in different habitat (founder effects), which eventually reduces the genetic variation (Willi, Fracassetti, Zoller & Van Buskirk, 2018) and increases the population size. As a result, the individuals within the same species (and population) have low nucleotide diversity, but high rate of non-synonymous mutations.

5.2. Molecular evolution of Australian *Eugongylinae* skinks

There are two commonly-used hypothesis that explain variation in mutation rate (Gillooly et al., 2004): metabolic rate and generation time. According to metabolic rate hypothesis, metabolism produces oxygen radicals which might result in DNA damage (Martin & Palumbi, 1993). This will prompt the body to repair the damage, and mutations might arise due to incorrect repair. According to Gillooly et al. (2001), temperature is positively correlated with metabolic rate of organisms, including reptiles. This means that higher temperature will result in higher metabolic rate, and thus mutation rates. Our results support this hypothesis, where species that live in higher temperature have significantly higher substitution rates than the ones in colder habitat, given that synonymous substitutions are nearly neutral and thus synonymous substitution rate approximates mutation rate. Even though the results of other variables are still significant after controlling for temperature, we find temperature as the strongest predictor for variation in both dN and dS, which is also found in previous study (Garcia-Porta et al., 2019). Alternatively, Husby et al. (2011) and Berger et al. (2018) highlighted the correlation between the strength of selection and mutation rates. Elevated temperature is proposed to increase the strength of selection, introducing adverse (and potentially novel) condition. This, in turn, results in mutational load and variance that affect the species fitness. However, this hypothesis is not reflected in our results, as temperature does not only explain the variation in dN, but also dS.

On the other hand, the link between temperature and substitution rates can be indirectly due to the association between metabolic rate and life-history traits. One mechanism is generation time hypothesis (Gillooly et al., 2004), in which species with shorter lifespan (and higher metabolic rate) undergo more DNA replication per unit of time, so they will have higher mutation rates than long-living ones. However, Stark et al. (2018) showed that temperature has significant marginal effect on longevity in Squamates, and no relationship on reptiles in general. The global application of this hypothesis has also been challenged by results from other studies that compared different

lineages (Avice et al., 1992; Martin et al., 1992). This suggests that generation time (i.e. mean age of breeding females) is not likely the main predictors of substitution rates.

Alternatively, other life-history traits, such as body size and fecundity (Bromham, 2011), have been proposed to be the predictors of mutation rates. These traits are also correlated with metabolic rate. Body size has long been associated with mutation rate in vertebrates, coupled with generation times (Martin & Palumbi, 1993; Bromham, 2002). The idea is that larger species will have higher chance of accumulating mutations in both germline and somatic cells due to more genome copies per generation. As a result, large-bodied organisms invest heavily in DNA repair, which becomes one of the reasons why they tend to live long (Promislow, 1994).

The negative association between body size and molecular evolution rate has been identified in mammals, birds, and reptiles (Bromham et al., 1996; Mooers & Harvey, 1994; Bromham, 2002). In this study, body size explains around 7% variation in both dN and dS that cannot be explained by temperature. Even though its explanatory power is lower than temperature, it should not be overlooked as it still has statistically significant negative correlation with substitution rates after controlling for temperature.

One mechanism for the link between body size and substitution rate that is unlikely to be confounded by temperature is through population size. The bigger the body size, the smaller the population size will be as each individual requires more resources in limited spaces. In turns, small population size will limit the strength of purifying selection, allowing more mutations to be fixed within the population. However, our results show that temperature and other variables have similar contribution for both dN and dS, while population size-related traits (e.g. heterozygosity, geographic range, niche breadth) are also very weak predictors. Thus, population size is unlikely to be the main predictor here.

Another factor that might co-act with body size and generation time is fecundity. Bromham (2002) found that fecundity of reptiles positively correlates with mutation rate. Similar with generation time, higher fecundity might allow more mutations due to more frequent genome

replication (Britten, 1986). However, as explained in Bromham (2002), this is only true if the increase in gametogenesis includes more cell divisions in each production. Alternatively, fecundity might work with other predictors. For instance, Blueweiss et al. (1978) showed that the fecundity of reptiles and other amphibious/aquatic poikilotherms increases as its maternal body size increases. Other factors might be the mode-of-life of the species, including their habitat and biome (Healy et al., 2014). This association needs to be assessed further in future studies.

Chapter VI. Conclusion and Recommendation

This study shows that the population structure and diversity among *Morethia* skinks differ in relation to historical habitat stability and current climatic limits of species. In general, southern *Morethia* species have generally weaker phylogeographic structures than northern *M. storri* but not *M. ruficauda*, especially the AZ lineages. In accordance to our third hypothesis, *M. obscura* and *M. adalaidensis* have no phylogeographic structures, potentially due to their narrower climate and habitat distributions. In contrary, *M. boulengeri* has quite pronounced phylogeographic structure which might be related to the presence of lake and rivers in eastern Australia. This warrants further evaluation of species boundaries with expanded sampling. Moreover, *M. obscura* (and potentially *M. adalaidensis*) also shows recent rapid population growth, similar with *M. ruficauda*. However, their timings are different: *M. obscura* experienced more recent population expansion than *M. ruficauda*. This might due to more sustained aridification in southern part of Australia, which forced the organisms to contract to refugia and expanded later.

Furthermore, our results also suggest the role of temperature as strong predictors of dN and dS among Eugongylinae skinks, where species that live in hotter temperature tend to have higher mutation rates than the ones in colder habitat. Specifically, mean AMT explains 35% and 40% of variations in dN and dS, respectively, and about 20% of variations in both after controlling for other factors. While the other variables explain less than 10% of variation in mutation rates after controlling for temperature, they still hold a potential to be confounding factors as their associations are still statistically significant. Future studies should consider and assess these possibilities to produce more reliable and informative results.

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